Fire Protection in Refineries

API RECOMMENDED PRACTICE 2001
SEVENTH EDITION, JUNE 1998

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Fire Protection in Refineries

Health & Environmental Affairs Department
Safety and Fire Protection Subcommittee

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FOREWORD

The term fire protection used in this publication includes measures taken to prevent fires, as well as those to minimize, control, or extinguish fires already burning. A thorough approach to fire protection starts with an understanding of the ignition and combustion processes, including control of potential fuel sources with an emphasis on containment. This publication gives some basic information on these subjects and identifies sources of more detailed information. While sections of this document discuss general design principles, it is not intended as a design manual. Rather, it presents guidance for those providing fire protection services to refineries and gives reference to sources of more detailed, design-related information.

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Fire Protection in Refineries

1 General

1.1 SCOPE

The purpose of this publication is to provide a better understanding of fire protection problems and the steps needed to promote the safe storage, handling, and processing of petroleum and petroleum products in refineries and the safe shipment of these products.

1.2 DEFINITIONS

Some of the terms used in this document are defined in sections 1.2.1 through 1.2.8. Additional definitions and an expanded discussion of fire-related hydrocarbon properties and phenomena are included in Section 2.

1.2.1 combustible: As defined by the National Fire Protection Association (NFPA) and used in this document, it refers to any solid that can burn or to any liquid that has a flashpoint of 100°F (37.8°C) or greater. See Section 2.3 and NFPA 30 for further explanation noting alternate classification systems for combustible liquids.

1.2.2 combustion (burning): The rapid reaction of oxidizable material with an oxidizer, usually oxygen from the air, followed by the development of heat. This reaction usually produces flames.

1.2.3 flammable: As defined by NFPA and used in this document, it refers to any gas that can burn or to any liquid that has a flashpoint below 100°F (37.8°C). The archaic term *inflammable* is obsolete. See Section 2.3 and NFPA 30 for a further explanation noting alternate classification systems for flammable liquids.

1.2.4 fuel: Material capable of undergoing combustion. It is the material which burns to feed a fire.

1.2.5 hazard: An inherent chemical or physical property with the potential to do harm (toxicity, corrosivity, stored chemical, or mechanical energy).

1.2.6 process: The refinery equipment, vessels, and piping in which refining takes place to "process" crude oil into manufactured products.

1.2.7 risk: The probability of exposure to a hazard which results in harm.

1.2.8 switch loading: Loading of low-vapor-pressure (high-flash-point) materials into containers where flammable vapors may be present from previous use, such as when diesel fuel is loaded into a tank which last carried a cargo of gasoline. API RP 2003 provides additional information on the static ignition hazards associated with switch loading.

1.3 REFERENCED PUBLICATIONS

The most recent editions of each of the following standards, codes, and publications are referenced in this recommended practice as useful sources of additional information. Additional information may be available from the cited Internet World Wide Web sites.

API	www.api.org
RP500	*Classification of Locations for Electrical Installations at Petroleum Facilities*
API510	*Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration*
RP520	*Sizing, Selection and Installation of Pressure-Relieving Systems in Refineries, Part I Sizing and Selection, and Part II Installation*
RP521	*Guide for Pressure-Relieving and Depressuring Systems*
API560	*Fired Heaters for General Refinery Services*
API570	*Piping Inspection Code: Inspection, Repair, Alteration, and Rerating of In-Service Piping Systems*
RP572	*Inspection of Pressure Vessels*
RP573	*Inspection of Fired Boilers and Heaters*
RP574	*Inspection of Piping, Tubing, Valves, and Fittings*
RP575	*Inspection of Atmospheric and Low-Pressure Storage Tanks*
RP576	*Inspection of Pressure Relieving Devices*
Std610	*Centrifugal Pumps for General Refinery Service*
Std617	*Centrifugal Compressors for General Refinery Service*
Std618	*Reciprocating Compressors for General Refinery Services*
Std620	*Design and Construction of Large, Welded, Low-Pressure Storage Tanks*
Std650	*Welded Steel Tanks for Oil Storage*
Publ653	*Tank Inspection, Repair, Alteration, and Reconstruction*
RP750	*Management of Process Hazards*
RP752	*Management of Hazards Associated with Location of Process Plant Buildings*
Std2000	*Venting Atmospheric and Low-Pressure Storage Tanks: Nonrefrigerated and Refrigerated*
RP2003	*Protection Against Ignitions Arising Out of Static, Lightning, and Stray Currents*
Publ2009	*Safe Welding and Cutting Practices in Refineries, Gasoline Plants, and Petrochemical Plants*

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Publ2021 *Fighting Fires In and Around Flammable and Combustible Liquid Atmospheric Petroleum Storage Tanks*
Publ2030 *Application of Fixed Water Spray System for Fire Protection in the Petroleum Industry*
RP2201 *Procedures for Welding or Hot Tapping on Equipment In Service*
Publ2210 *Flame Arresters for Vents of Tanks Storing Petroleum Products*
Publ2214 *Spark Ignition Properties of Hand Tools*
Publ2216 *Ignition Risk of Hot Surfaces in Open Air*
Publ2218 *Fireproofing Practices in Petroleum and Petrochemical Processing Plants*
Publ2219 *Safe Operating Guidelines for Vacuum Trucks in Petroleum Service*
RP2350 *Overfill Protection for Petroleum Storage Tanks*
Std2510 *Design and Construction of LP-Gas Installations at Marine and Pipeline Terminals, Natural Gas Processing Plants, Refineries, and Tank Farms*
Publ 2510A *Fire-Protection Considerations for the Design and Operation of Liquefied Petroleum Gas (LPG) Storage Facilities*
Std2610 *Design, Construction, Operation, Maintenance and Inspection of Terminal and Tank Facilities*

AIChE (CCPS)[1] *Guidelines for Engineering Design for Process Safety*
Guidelines for Evaluating the Characteristics of Vapor Cloud Explosions, Flash Fires and BLEVEs
Guidelines for Hazard Evaluation Procedures
Guidelines for Safe Automation of Chemical Processes
Guidelines for Technical Planning for On-Site Emergencies

ANSI[2,3]
B31.1 *Power Piping*
B31.2 *Fuel Gas Piping*
B31.3 *Chemical Plant and Petroleum Refinery Piping*
B31.4 *Liquid Transportation Systems for Hydrocarbons, Liquid Petroleum Gas, Anhydrous Ammonia, and Alcohols*
B31.5 *Refrigeration Piping*
B31.8 *Gas Transmission and Distribution Piping Systems*
B31.9 *Building Service Gas*
B31.11 *Slurry Transportation Piping Systems*
Z129.1 *Hazardous Industrial Chemicals—Precautionary Labeling*

ASME[3] *Boiler and Pressure Vessel Code*

ASTM[4]
D86 *Standard Test Method for Distillation of Petroleum Products*
D323 *Standard Method of Test for Vapor Pressure of Petroleum Products (Reid Method)*

Bureau of Mines[5]
Bull503 *Limits of Flammability of Gases and Vapors*
Bull627 *Flammability Characteristics of Combustible Gases and Vapors*

NFPA[6] *Fire Protection Handbook*
Flammable and Combustible Liquids Code Handbook
Furnace and Boiler Codes
10 *Portable Fire Extinguishers*
11 *Low Expansion Foam-Extinguishing Systems*
12 *Carbon Dioxide Extinguishing Systems*
12A *Halon 1301Fire Extinguishing Systems*
12B *Halon 1211Fire Extinguishing Systems*
13 *Installation of Sprinkler Systems*
15 *Water Spray Fixed Systems for Fire Protection*
16 *Installation of Deluge Foam Water Sprinkler Systems and Foam Water Spray Systems*
17 *Dry Chemical Extinguishing Systems*
20 *Installation of Centrifugal Fire Pumps*
24 *Installation of Private Fire Service Mains and Their Appurtenances*
25 *Water-Based Fire Protection Systems*
30 *Flammable and Combustible Liquids Code*
58 *Storage and Handling of Liquefied Petroleum Gases*
69 *Explosion Prevention Systems*
70 *National Electrical Code*
72 *National Fire Alarm Code*

[1]American Institute of Chemical Engineers Center for Chemical Process Safety, 345 East 47th Street, New York, New York 10017. www.aiche.org/docs/ccps.
[2]American National Standards Institute, 11 West 42nd Street, New York, New York 10036.www.ansi.org
[3]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.www.asme.org
[4]American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, Pennsylvania 19428. www.astm.org
[5]U.S. Bureau of Mines [part of NIOSH/CDC] Pittsburgh Research Laboratory, P.O. Box 18070, Pittsburgh, Pennsylvania 15236. www.cdc.gov/niosh/pubs.html
[6]National Fire Protection Association, 1 Batterymarch Park, Quincy, Massachusetts 02269.www.nfpa.org

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85C	*Furnace Explosions /Implosions in Multiple Burner Boiler-Furnaces*
325	*Fire Hazard Properties of Flammable Liquids, Gases, and Volatile Solids*
496	*Purged and Pressurized Enclosures for Electrical Equipment in Hazardous (Classified) Locations*
497A	*Classification of Class I Hazardous Locations for Electrical Installations in Chemical Process Areas*
497B	*Classification of Class II Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas*
600	*Industrial Fire Brigades*
780	*Lightning Protection Code*
325	*Fire Hazards Properties of Flammable Liquids, Gases and Volatile Solids*
1561	*Fire Department Incident Management System*
1962	*Care, Use and Service Testing of Fire Hose, Including Connections and Nozzles*
1964	*Spray Nozzles (Shutoff and Tip)*
2001	*Clean Agent Fire Extinguishing Systems*
OCIMF[7]	International Safety Guide for Oil Tankers and Terminals
OSHA[8]	
29 *CFR* 1910.38	*Employee Emergency Plans and Fire Prevention Plans*
29 *CFR* 1910.110	*Storage and handling of Liquefied Petroleum Gases*
29 *CFR* 1910.119	*Process Safety Management of Highly Hazardous Chemicals*
29 *CFR* 1910.120	*Hazardous Waste Operations and Emergency Response*
29 *CFR* 1910.147	*Control of Hazardous Energy (Lockout/Tagout)*
29 *CFR* 1910.156	*Subpart L—Fire Brigades*
29 *CFR* 1910.252	*Subpart Q—Welding, Cutting and Brazing*

2 Chemistry and Physics of Fire

There are inherent hazards associated with the processing, handling and/or storage of petroleum products, partly due to the volatility of many of these products. If proper precautions are not followed to prevent the concurrent presence of the elements of combustion there is the possibility of a fire or explosion which could result in risk of harm to exposed personnel, damage to equipment, and adverse effects on the environment.

[7]Oil Companies International Marine Forum, Bermuda, and International Chamber of Shipping, London.

[8]U.S. Department of Labor, Occupational Safety and Health Administration, 200 Constitution Avenue, N.W. Washington, D.C 20210. www.osha.gov

2.1 THE COMBUSTION PROCESS

Three components are necessary for a fire to begin—oxygen (usually from air) mixed in the proper proportions with fuel which has been exposed to sufficient heat to vaporize, with enough additional heat to initiate combustion and vaporize more fuel.

A fire starts and is sustained by a fourth essential component, a free-radical chemical chain reaction that enables the fire to continue as long as the three other components remain available. The four essential components are represented in Figure 1.

A basic understanding of the components of a fire and an appreciation of the related fire prevention measures will help reduce the risk of fire and aid fire personnel in their suppression efforts should a fire occur.

2.2 FUEL TYPES AND CLASSES

Fuels can be classified as three types: solids (coal, wood, plastic, etc); liquids (gasoline, crude oil, alcohol, etc.); and gases (propane, hydrogen, acetylene, etc.). As only gases burn, the combustion of a liquid or solid fuel requires partial conversion of the liquid or solid fuels into a gaseous state by heating. This process is called pyrolysis for solid fuels and vaporization for liquid fuels.

NFPA 10 classifies fires based on the fuel involved—either type A, B, C, or D. Fire extinguishing agents are often identified by this system based on the type of fire for which they are effective (i.e., a dry chemical fire extinguisher may carry a Class ABC rating where a pressurized water extinguisher carries only a Class A rating).

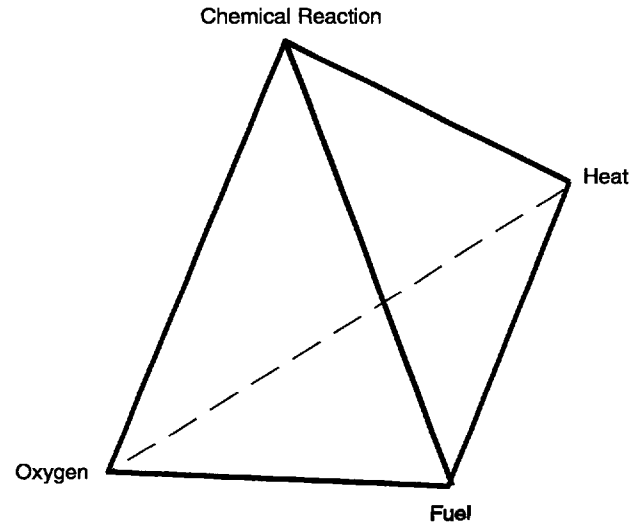


Figure 1—Fire Tetrahedron

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Class A fires are those involving ordinary combustible solid materials such as wood, coal, paper, rubber, and many plastics.

Class B fires are those involving flammable and combustible liquids and gases such as gasoline, crude oil, alcohols, LPG, and hydrogen.

Class C fires are those involving energized electrical equipment. While electricity is not a fuel, it represents a significant hazard to firefighters if improper extinguishing agents or methods are used. Once the electrical circuit is de-energized, the fire is then treated as a Class A or B, depending upon the fuel involved.

Class D fires are those involving combustible metals such as sodium, aluminum, magnesium, or titanium.

2.3 PROPERTIES OF PETROLEUM PRODUCTS

There are several physical properties exhibited by petroleum products which have a significant impact on their fire and explosion potential. These include the following:

Autoignition Temperature—Minimum temperature to which a fuel in air must be heated to start self-sustained combustion without a separate ignition source. This means that, should a leak occur on a line containing a petroleum product above its ignition temperature, ignition can occur independent of an ignition source.

Boiling Point—The temperature at which the vapor pressure of a liquid equals the surrounding atmospheric pressure. For purposes of defining the boiling point, atmospheric pressure shall be considered to be 14.7 psia (760 mm Hg). For mixtures that do not have a constant boiling point, the 20 percent evaporated point of a distillation performed in accordance with ASTM D86 shall be considered as the boiling point.

Fire Point—The temperature (usually a few degrees above the flash point) at which a liquid produces enough vapors to sustain combustion.

Flammable Range—A range of vapor-to-air ratios within which ignition can occur. The lower flammable limit (LFL) is the minimum vapor-to-air concentration below which ignition cannot occur. Atmospheres below the LFL are referred to as too lean to burn. The upper flammable limit (UFL) is the maximum vapor-to-air concentration above which ignition cannot occur. Atmospheres above the UFL are referred to as too rich to burn. Flammable ranges can vary widely, as illustrated by flammable vapor-to-air ranges for gasoline (1.4–7.6 percent) and acetylene (2.5–80 percent).

Flash Point—The lowest temperature at which a liquid gives off enough vapor to produce a flammable mixture with air immediately above the surface. A source of ignition is needed for flash to occur. When this temperature is above ambient, vapors will ignite but will not continue to burn until heated to the "fire point." The flash point temperature can be very low for volatile petroleum products [for instance, the flash point for gasoline is typically quoted as about –45°F (–43°C)].

Specific Gravity—ratio of the weight of a given substance to the weight of an equal volume of a standard substance (water for liquids and air for gases). This is frequently referred to as vapor density for gases. Since the specific gravity of the standard equals one, liquids with a specific gravity less than one will float on water (unless they are water soluble like most alcohols). Most liquid petroleum products have specific gravities less than one. Likewise, gases with a specific gravity of less than one (e.g., hydrogen and methane) will rise in the atmosphere, whereas gases with a specific gravity greater than one (e.g., almost all hydrocarbons with two or more carbon atoms such as ethane, propane and butane) will tend to stay close to the ground and seek low-lying areas.

Vapor Pressure—The pressure exerted by the vapor of a substance when the substance and its vapor are in equilibrium. Equilibrium is established when the rate of evaporation of a substance is equal to the rate of condensation of its vapor. In general terms, the higher the measured laboratory vapor pressure the more likely that liquid is to give off vapors under "real world" conditions. Reid Vapor Pressure, a measure commonly used to characterize gasoline and other hydrocarbon volatility, is measured at 100°F (37.8°C) in a closed container.

The importance of these basic vapor generation and combustion properties can be seen in the way flammable and combustible liquids are classified (see the following discussion and NFPA 30 for more information):

Flammable liquids have flash points below 100°F (37.8°C) and vapor pressures not exceeding 40 psia (2068.6 mm Hg) at 100°F (37.8°C). Liquids with vapor pressures above 40 psia (276 kPa) at 100°F (37.8°C) are considered gases by NFPA.

a. Class IA—Flash point below 73°F (22.8°C) and boiling point below 100°F (37.8°C).

b. Class IB—Flash point below 73°F (22.8°C) and boiling point above 100°F (37.8°C).

c. Class IC—Flash point at or above 73°F (22.8°C) and below 100°F (37.8°C).

Combustible liquids have flash points at or above 100°F (37.8°C).

a. Class II—Flash point at or above 100°F (37.8°C) and below 140°F (60°C).

b. Class IIIA—Flash point at or above 140°F (60°C) and below 200°F (93°C).

c. Class IIIB—Flash point at or above 200°F (93°C).

OSHA uses NFPA definitions for flammable and combustible. Alternate systems using 140°F (60°C) as the dividing point between flammable and combustible appear in ANSI/CMA Z129.1-1994 and the regulations of the U.S. Department

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of Transportation and the United Nations. The NFPA classification system is used in this document and is widely used for facility-based fire protection purposes in the United States. For regulatory compliance purposes (such as labeling for off-site transportation), reference should be made to the specific regulations or codes governing the activity of concern.

2.4 OXYGEN

Oxygen is readily available as air contains about 21 percent oxygen under normal circumstances. However, as it relates to hydrocarbon products processed and stored in closed systems, often there is insufficient oxygen (minimum of 10 to 12 percent required) to support combustion unless a leak occurs. Some materials (e.g., ammonium nitrate, trinitrotoluene) contain sufficient oxygen in their chemical composition to support combustion even in an inert atmosphere.

2.5 HEAT (SOURCE OF IGNITION)

In the petroleum industry, oxygen is always available from the ambient environment. If leaks occur despite extensive efforts to prevent them, there is the potential for a combustible mixture of fuel and air to exist. Therefore, emphasis is placed on limiting sources of ignition. An attractive approach would be the strict exclusion of all possible ignition sources from areas in which flammable substances could be present. However, strict application of this principle is impractical (for instance furnaces are often needed to provide process heat).

The five general categories of heat energy are chemical, electrical, mechanical, nuclear, and solar. Those heat energies applicable to petroleum products are discussed in the following sections.

2.5.1 Chemical

Heat of Combustion—The heat energy, in the form of a flame, that is released by the chemical reaction of a fire.

Spontaneous Heating—The heating of an organic substance without the addition of external heat. A common example is oily rags which slowly oxidize releasing heat. In the petroleum industry, spontaneous heating can occur from pyrophoric iron sulfide deposits. These deposits may form in equipment handling petroleum products that contain hydrogen sulfide or other sulfur compounds when there is insufficient oxygen to fully convert the sulfur compounds to sulfates. Pyrophoric iron sulfide deposits are generally not a problem if they stay wet or remain in an oxygen-deficient environment. However, once exposed to air, they can oxidize and heat, providing a source of spontaneous ignition if flammable petroleum vapors are also present.

Heat of Solution and Heat of Dilution—Represent the heat released by dissolving matter in liquid or the mixing of unlike liquids. While not an ignition source, substantial quantities of heat can be generated creating a potential fire hazard in certain circumstances. The heating of caustic (sodium hydroxide) or acid process materials can raise temperatures, vaporize liquid materials and result in container failure and releases of hydrocarbon providing accessible fuel for a fire.

2.5.2 Electrical

Arcs and Sparks—The energy released by current flowing across an air gap can cause significant heating of the air and of the electrodes (equipment) across which the arc or spark travels. Arcing can occur when current flow is initiated or interrupted by opening or closing a switch or by the intermittent contact of loose connections. Sparks and arcs from normal AC or DC refinery equipment are assumed to provide sources of ignition and treated with appropriate precautions.

Static Electricity—The buildup of a positive charge on one surface and negative charge on another is a natural phenomenon associated with the flow of fluids (such as hydrocarbons) through pipes. The charges are attracted to each other and a spark can occur when the voltage differential is sufficient to ionize an air gap and provide a path to transfer electrons between the surfaces to become evenly charged again. Grounding and bonding are used during petroleum transfer operations to provide a conductive path through which accumulated static charges on the hydrocarbon surface can recombine with opposite charges on the vessel shell or loading arms to prevent sparking and potential ignition.

Lightning—Static electricity on a very large scale. Protection against direct lightning strikes is difficult. The floating roofs of storage tanks use shunts to metallic shoes sliding on the grounded tank shell to provide a path to ground as protection from induced charge sparks caused by lightning.

See API RP 2003, *Protection Against Ignitions Arising Out of Static, Lightning, and Stray Currents*, for more in-depth information.

2.5.3 Mechanical

Friction—Heat created by the rubbing of two surfaces against each other. High-energy friction sparks, such as those produced by failure of pump bearings leading to rubbing of mechanical seals, can ignite flammable mixtures. Typical low energy sparks produced by the impact of steel, stone, or other hard substances (including hand tools) will not ordinarily ignite petroleum vapors. See API Publication 2214 for further information.

2.5.4 Hot Surfaces

Although not a type of heat, hot surfaces can be a source of ignition if they are large enough and hot enough. The hot surface vaporizes the liquid, a flammable atmosphere forms, and the hot surface raises the flammable atmosphere to its autoignition temperature. Generally, the smaller the heated surface, the hotter it must be to provide a source of ignition. The heat flow

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from the hot surface must overcome the cooling effect of the evaporating liquid or expanding vapor while providing enough energy to heat the vapor to autoignition temperature. As a result, the larger the heated surface in relation to the quantity of vapor-to-air mixture, the more rapidly ignition will take place and the lower the surface temperature necessary for ignition. Because of the heat absorbed by vaporization, it is usually more difficult for a hot surface to ignite flammable liquids than to ignite their vapors. Less-volatile combustible liquids which evaporate at higher temperatures can be ignited more readily by contact with hot surfaces, as evidenced by the burning of motor oil splashed on a hot automobile exhaust manifold where gasoline may vaporize without ignition. See API Publication 2216 for additional information.

2.6 SPECIAL SITUATIONS

The inherent properties of hydrocarbon materials processed and produced in refineries present several unique hazards related to hydrocarbon storage. Some of these phenomena have the potential to cause substantial damage. These should be addressed while designing for fire prevention and planning for emergency response. Operational precautions, such as outlined in this recommended practice, and well-planned emergency response can reduce the likelihood of occurrence and help prevent escalation in the event of an incident.

Boilover—Sudden overflow or ejection of the contents of an oil storage tank (crude oil or certain other viscous liquids) during a full surface fire due to a heat wave (layer of hot, heavy oil) reaching water or water-oil emulsion at the bottom of the tank. The water flash-boils and turns to steam with rapid expansion which can send the tank contents a significant distance. Boilover occurs only with tanks containing oils with a wide boiling range, including both a viscous fraction and light ends (e.g., like crude oil, but not gasoline). In extreme cases, substantial amounts of flammable liquids can be expelled creating a serious hazard for hundreds of feet surrounding the tank. See the NFPA *Flammable and Combustible Liquids Code Handbook* for more information.

Slopover—Results from a water stream being applied to the hot surface of a boiling oil, when the oil is viscous and its temperature exceeds the boiling point of water.

Frothover—Overflowing a container not on fire when water boils under the surface of a viscous hot oil. This is one reason product rundown temperatures to tankage should be monitored and controlled.

BLEVE (Boiling Liquid Expanding Vapor Explosion)—Catastrophic failure of a container into two or more major pieces at a temperature well above the boiling point of the contents at atmospheric pressure. Classic cases of BLEVE have involved liquefied petroleum gas containers, such as LPG rail cars, where violent fire and explosion has been caused by rupture of the vessel wall. This rupture can be caused by localized overheating (such as by jet fire impingement), which reduces the strength of the container while increasing the internal pressure beyond the venting capability of the vessel being exposed to fire. The phenomenon is applicable to volatile liquids as well as to LPG. (See the NFPA *Flammable and Combustible Liquids Code Handbook* for further information.)

More information on the hazards involved with petroleum products and the fire prevention/protection measures available to deal effectively with these hazards can be found in other sections of this publication and in cited references.

3 Refinery Design

3.1 GENERAL

This section discusses general design principles for refineries, with emphasis on fire prevention and fire mitigation. This recommended practice is not a design manual. Many of the references cited can provide specific design information. The principles outlined are intended as guides to good engineering practice. It is not intended that the fire protection systems and equipment designs discussed be applied retroactively to existing installations. The principles described identify certain areas to consider during refinery design for new construction, or upgrades and expansions, to help reduce the possibility of a fire and to mitigate fire damage. Variance from the principles described does not necessarily imply that a refinery is inadequately protected against fire. Safety and fire protection practices must be evaluated within the design philosophy of the specific project involved, on the basis of experience, accepted process safety management practices, normal industry practice, and regulatory requirements.

3.2 HAZARD ANALYSIS

Preventing and mitigating loss of containment of flammable and combustible materials is important to the safety of a refinery. Systematic hazard analysis with implementation of the findings can help a refinery avoid major releases of process materials and reduce the risk of potential fires. These activities can also help the facility to implement the Management of Process Hazards as outlined in API RP 750 and to satisfy regulatory requirements (OSHA 29 *CFR* 1910.119).

The use of hazard analysis during process design, site selection, refinery layout, equipment selection, and civil and structural design will aid in the mitigation of hazards which might result in fires. Safety elements should be considered as early as the conceptual stages and continue throughout all stages of process design for new construction, upgrades, and expansions. The hazard analysis should include input from research and development personnel, designers, and refinery management. The purpose of the hazards analysis is to identify early in the design process any potential hazards inherent

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in a process design, and to integrate protective measures to address these potential hazards.

The analysis should include an evaluation of the fire, internal explosion, and reactivity potential of process equipment and contents and the consequences associated with such events.

There are several types of hazard analysis methodology currently in use. Some of the more prevalent types are Hazard and Operability Studies (HAZOP), Failure Mode and Effects Analysis (FMEA), and Fault Tree Analysis. See API 750, *Management of Process Hazards*; the AIChE-CCPS book, *Guidelines for Hazard Evaluation Procedures*; and OSHA 29 *CFR* 1910.119 for additional information on these methodologies and their use.

3.3 PROCESS DESIGN

Process design includes evaluation of the properties of process materials, process conditions, and inventories. Process design should emphasize the principles of "inherent safety." For example, minimization of hazardous material inventories in the process design phase can be effective in decreasing the consequences of a release due to leaks or explosions.

It is important to have basic understanding of (a) general properties, such as reactivity, flammability, toxicity, and stability of materials used in the process, and (b) procedures for safe handling of stored materials used in the process.

In reviewing process conditions, it is important to understand the operating parameters—such as pressure and temperature—intended by the design, along with their associated margins of safety. Potential hazards are introduced by abnormal operating conditions which exceed process design conditions because extremes will increase stress and may cause undesired chemical reactions or equipment failure.

3.4 EQUIPMENT DESIGN

Proper design and arrangement of equipment in processing units, storage areas, and loading/unloading areas can prevent potential hazards from resulting in fires. Design should be consistent with accepted industry codes and supplemented with good engineering practices by personnel having a knowledge of equipment service and its potential hazards. Key aspects of the design should include construction materials, mechanical design and equipment construction, and the process control system. The following lists are examples (but are not exclusive of other conditions or mechanisms).

The following conditions can lead to fires or explosions:

a. Vapor clouds resulting from release of flammable liquids or gases.
b. Spill releases of flammable liquids.
c. Pressure increase in vessels beyond their design.
d. Loss of inerting, enriching, or diluting systems.
e. Increase in temperatures due to unstable conditions.
f. Formation of flammable mixtures inside equipment.
g. Mixing of incompatible materials.
h. Conditions resulting in dust explosions.
i. Undue vibration and shock from process conditions.

The following are potential mechanisms for material release from process equipment:

a. Vessel rupture due to pressure/flow demand greater than relief capacity.
b. Vessel rupture due to brittle fracture.
c. Failures of flanges, gaskets, seals, or plugs.
d. Weld or casting failures.
e. Overstressing of vessel shell.
f. Excess vibration.
g. Excess corrosion/erosion.
h. Failure due to external loading or impact.
i. Internal explosion.
j. Tube rupture from overheating.

3.4.1 Materials of Construction

The suitability for the intended use of materials at a refinery has a bearing on fire loss risk. The standards, specifications, and recommendations of nationally-recognized authorities should be consulted and applied where relevant. In many instances, specifications have been promulgated by the American Petroleum Institute to meet the particular needs and requirements of oil refineries. Additional information is available in the publications of such organizations as the American Society for Testing and Materials (ASTM); the American Society of Mechanical Engineers (ASME); the American National Standards Institute (ANSI); the National Fire Protection Association (NFPA); ASM International (formerly the American Society of Metals);[9] and Underwriters Laboratories Inc. (UL).[10]

A careful review and selection of materials by qualified personnel decreases the potential for materials failure. Proper selection of the materials of construction requires a thorough knowledge of the internal process, the exterior environment, failure modes, correct material application and fabrication techniques, maintenance philosophy, and inspection intervals. The current trend toward positive material identification programs provides a quality-control method for preventing the unintended use of inappropriate materials, helping to avert one potential cause of material failures.

Corrosion is a recurring cause of loss of containment of flammable and combustible materials which leads to fires. The potential for unexpected corrosion is therefore one of the most important factors in selecting the construction materials

[9]The Materials Information Society, 9639 Kinsman Road, Materials Park, Ohio 44073-0002. www.asm-intl.org
[10]Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, Illinois 60062. www.ul.com

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and maintaining surveillance programs. See the relevant API publications for inspection of refinery equipment.

The basis for selection of materials is the performance of the materials under process design conditions and the interaction of the materials with the external environment. Operating conditions include startup, shutdown, and upset conditions. Any potential problem anticipated with a particular material should be discussed during hazard analysis sessions. It is important that personnel with relevant expertise be consulted during materials selection.

3.4.2 Mechanical Design and Equipment Construction

Refinery operations frequently use high pressures and temperatures that put stress on equipment. Equipment used in the process must be designed to withstand the stresses of the operating conditions to which it will be subjected.

In addition to a quality control program for materials, a sound quality control program for equipment construction is important and should be in place for construction activities to verify that construction is in conformance with design specifications.

3.4.2.1 Pressure Vessels

Design and construction specifications for vessels subject to pressures of 15 pounds per square inch gauge or more are given in Section VIII, "Unfired Pressure Vessels," of the ASME *Boiler and Pressure Vessel Code*.

Suitable provisions must be made for cleaning and ventilating a vessel when it is internally inspected. It should be isolated from other equipment and from flammable, toxic, or inert materials.

Reactors, columns, exchangers, and boilers are typical pressure vessels found in a refinery which require special design considerations for structural supports, instrumentation, and protective systems.

3.4.2.1.1 Internal Design

Internal design of vessels should avoid pockets where the lack of drainage would allow water to accumulate, particularly during startup. The sudden generation of steam caused by the contact of hot-oil charge stocks with water can result in dangerous overpressure of equipment.

3.4.2.1.2 Protection of Vessels

Vessels with internal refractory or insulation, or vessels which handle materials entirely in the vapor phase, may be subject to rapid overheating and rupturing when exposed to fire if fire protection cooling, insulation, or relief is not provided.

3.4.2.1.3 Flanges

The vessel connections should be flanged as close to the vessel as practical to permit installation of blinds. During the design and installation stages consideration should be given to access to the flanges for blinding.

3.4.2.2 Block Valves

Where large volumes of flammable liquid are contained in process vessels, block valves should be installed on connections below the liquid level. This will permit shutoff of fuel flow if downstream piping or equipment becomes involved in a fire.

3.4.2.3 Isolation Valves

Consideration should be given to access to valves during fire conditions. Adequate equipment isolation valves should be provided within process unit areas to isolate equipment during fire situations and to permit blinding for maintenance and inspection. During blinding operations, blinds suitable for equipment-rated pressure should be installed. Isolation valves and drains should be provided for equipment that may be opened or removed during repair operations. The use of automatic (fire- or heat-actuated) self-closing valves is not recommended unless confirmed as inherently safe by a rigorous process safety review—since in the event of a fire, these valves could close and prevent the orderly shutdown of equipment, or transfer of product from tanks or vessels, during an emergency.

3.4.2.4 Oil and Gas Piping

Specifications for the construction of oil and gas piping at refineries are contained in ANSI B31.3. The piping layout should be designed with battery limit manifolds so any major processing area within the refinery can be blocked off in case of fire. This can help prevent the flow of fuel into the fire area and reduce disruption of operations in other areas. Section 1.3 provides references to other piping guidelines from ANSI.

During the design stages of a refinery project the process information, such as type of fluids, temperatures, pressures, and flow conditions, must be detailed so that compatible gaskets, pipe construction materials, line flange classes, and pipe wall thicknesses can be provided for the piping system. Special piping and valving may be needed to address two-phase flow, high pressure drops, corrosive or erosive fluid properties, or high-velocity flow conditions.

Other special design features of piping which may be of concern and need additional discussion and detail include the following:

a. Thermoplastic or plastic lined pipe.
b. Double-walled pipe.
c. Above-ground versus below-ground service.

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d. Cathodic protection and grounding features.
e. Jacketed or heated piping.
f. Corrosion under insulation and supports on piping.
g. Special velocity effects (corrosion, erosion, vibration, noise, water hammer, and static electricity.
h. Special coatings and insulation.
i. Dead-leg potential for water accumulation and freezing or corrosion.
j. Piping specification breaks.

Consideration should be given to minimization of unshielded flangeless fittings, victaulic couplings, or long bolt flanges. When exposed to fire, long bolts can expand and allow flange connections to loosen.

Failure of small piping connections to mechanical equipment has caused numerous refinery fires. However, adequate bleeders and vents must be provided to permit removal of water from the system during startup; to provide adequate purging of air from the system; and to allow verification that out-of-service equipment has been drained and depressured.

Valves on these small connections should be located as close as possible to the takeoff points and sized to provide resistance to vibration and accidental breakage. Where vibration is a concern, suitable bracing should be provided and all-welded construction should be considered. Where screwed connections are used, backwelding up to the first block valve may be desirable. Small-diameter drain piping associated with valves should incorporate threaded plugs as a safeguard against leaks.

3.4.2.5 Purging Systems

In the design of processing equipment, an inert purging system may be desirable. This system prevents flammable mixtures from being present by excluding air from equipment during unit startups and by purging equipment of hydrocarbons prior to maintenance and repair work. Steam, nitrogen, carbon dioxide, and flue gas are common materials used for purging. Work practice procedures should address the inherent oxygen deficiency hazards of inert gases to provide personnel safety.

Purging facilities should be installed on fired heaters and other fired equipment to ensure removal of flammable vapors from the firebox before lightoff. Proper purging procedures should be developed and followed, since lack of sufficient purging can be a contributing cause of fired heater and boiler explosions. Verification of adequate purge flow provides a safeguard of key importance.

3.4.2.6 Pumps and Compressors

Proper design of pumps and compressors, along with proper installation in conformance to design, can significantly minimize the chance of process area fires. Reference should be made to API Standards 610, 617, and 618.

An operating compressor can be seriously damaged by liquid carryover. To prevent such carryover, suction and interstage knockout drums should be provided. High-level alarms and shutdown devices may also be needed.

The design of compressor systems should provide for the removal of air. Vents should be provided on the distance pieces of reciprocating compressors to avoid blowing gas into the compressor crankcase in the event of packing leakage. Vents should be located so as to eliminate the potential for pockets which would not allow all air to be expelled from the system. The flammability and toxicity of the released gases will determine the safest location for the discharge point of such vents.

Drainage from suction and discharge bottles, suction knockout drums, distance piece drains, and gas engine fuel gas knockout drums should be routed to safe disposal points. Process bypasses on compressor systems require special design considerations, such as high-temperature alarms, shutdowns, and recycle coolers for over-temperature conditions resulting from recycling. Vibration should be monitored for major pumps and compressors. For critical equipment, consideration should also be given to providing automatic and/or remote shut-down ability.

Small screwed connections to pumps and compressors are prone to fatigue and should be eliminated, where possible, or seal-welded if elimination is not possible.

3.4.2.7 Gas-Fueled Engines

Generally, gas-fueled engines are provided with automatic shutdown controls based on low lube oil pressure, high jacket water temperatures, and engine overspeed. In some instances, shutdown controls are provided in cases of vibration or of a high liquid level in the suction knockout drum.

The location of the air intake to gas engines should be configured to minimize the possibility of pulling in hydrocarbon vapors. To prevent the entry of fuel gas, the air piping should be vented near the engine when it is not in use. There have been recorded incidents in which starting air system explosions resulted from accumulation of lubricating oil in the air piping receiver and from the backup of fuel gas from the engine.

3.4.2.8 Fired Equipment

API Standard 560 and NFPA 85C should be used as references in the design of fired heaters and boilers. Fired equipment, such as process heaters and boilers, has the potential for mechanical equipment failure due to their harsh operating conditions. The following should be considered in the design of this equipment:

a. Fired equipment provides ignition sources and should be located on the periphery of the unit, with additional spacing from process equipment.
b. Fuel systems should be designed for positive isolation when the equipment is shut down, to prevent fuel accumulation in the firebox and a possible explosion on start-up.

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c. The firebox should be designed with air, nitrogen, or steam purging so that any accumulated hydrocarbon vapors can be purged from the firebox before start-up.

d. The fuel system should be designed to prevent flameout from temporary loss of fuel through the use of pilots, minimum flow bypasses, or automatic shutdowns.

e. There should be instrumentation to detect hazardous situations (i.e., loss of process flow, low fuel pressure, loss of combustion air, etc.) and to shut down fuel to the furnace as required. A shut-down system should be separate from the control system.

f. Surface drainage should be designed to prevent flow of liquid from the furnace toward other process equipment and from other process equipment to the furnace.

g. Remote isolation should be considered for process streams being heated in fired heaters.

3.4.2.9 Utility Systems

The contamination of utility systems, such as steam, air, water, fuel gas, and inert gas could result in serious safety problems. The most probable sources of contamination are utility connections joined to process equipment. Permanent connections should be avoided whenever possible. Water lines coming from cooling light hydrocarbon service should have vent headers installed for surge protection and for allowing venting of light hydrocarbon contamination prior to entry of the water stream into cooling water towers. Other means of surge protection may also be used.

If the utility is used in processing, a check valve and a block valve (preferably "double-block-and-bleed") should be installed at the connection. This arrangement can reduce the likelihood of a backflow of hydrocarbon if utility pressure is lost.

If the utility is required only when the equipment is out of service, such as for flushing, purging, or regeneration of catalyst, the utility connections should be blinded or disconnected. Should the utility be provided for intermittent or emergency use, double block valves with open bleeders and a check valve should be installed. All utility connections should be adequately identified. Check valves should be used at multi-service utility "hose station" headers.

Plant air systems have experienced fires from the accumulation of lube oil. Air systems should be designed to permit periodic cleaning to remove accumulated lube oil. The use of fire-resistant lubricants in air compressor crankcases should be considered. In new designs it may be possible to use oil-free compressors. Proper design and maintenance can eliminate excessive discharge temperatures caused by the air compressor and by fouling of the air cooling system. Instrument and plant air systems might become contaminated with hydrocarbon if the compressor air intakes are located too close to potential sources of hydrocarbon releases; the location of air intakes can be addressed in process safety reviews.

If fuel gas is burned in fired heaters, suitable knockout facilities (and line heat tracing where climate dictates) should be provided to prevent the carryover of liquid condensate into burners. Fuel line block valves should be located at a sufficient distance from fired heaters to permit remote shutdown in case of fire.

3.4.2.10 Loading Racks

Historically, a leading cause of loading rack fires has been overfilling. The most important aspect of controlling loading rack fires is to stop product flow to the rack. Because valves at the loading manifold may be inaccessible under certain fire conditions, there should be facilities which can block off the product lines to the rack from a remote location. These facilities may include remote pump shutdown; remotely operated block valves in the product lines; block valves in the lines at a short distance from the rack that can be readily reached by the operator; and alarms or other communications which can notify other personnel to expeditiously block out the rack.

In addition, static electricity may be an ignition source when materials are loaded at temperatures above their flash points or when "switch loading" occurs. Normal precautions include electrical bonding to facilitate relaxation of static charges, procedural control of fill rates and techniques (such as avoiding splash filling, bottom loading, and continuity of fill lines), and review of facilities (such as filtration) to minimize static charge build-up. Refer to API RP 2003 for a discussion of static electricity phenomenon.

3.4.2.11 Storage Tanks

Design and management of storage tank facilities has been addressed by API and other industry bodies in a number of standards. API Std 2610, *Design, Construction, Operation, Maintenance and Inspection of Terminal and Tank Facilities*, discusses design and operation of tank facilities. Along with environmental concerns, a major consideration in the design of storage tank installations is reducing fire risk. Risk reduction methods include: control of spills and protection against overfill (API RP 2350, *Overfill Protection for Petroleum Storage Tanks*); maintenance of tank integrity (API Publ 653, *Tank Inspection, Repair, Alteration, and Reconstruction*); proper arrangement and spacing of tanks (NFPA 30); and providing fire control and extinguishment equipment and systems (API 2021, *Fighting Fires In and Around Flammable and Combustible Liquid Atmospheric Petroleum Storage Tanks.*)

Typical storage areas may contain atmospheric, pressurized, refrigerated, or heated tankage. Additional API and NFPA standards listed in Section 1.3 address these more-specialized storage facilities. Typically, large scale bulk storage is located in adjacent tank farms, with smaller storage facili-

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ties distributed throughout various areas within process batteries.

Detailed information on mechanical design, fabrication, and nondestructive examination of storage tanks, and protective systems can be found in API Standards 620 and 650. It is not within the scope of this document to discuss issues which are covered in detail by the referenced standards.

3.4.3 Process Control Systems

The design and methods used to provide control, as well as the accuracy and reliability of the instrumentation, affect the safe operation of the refinery. Reliable measurement of pressure, temperature, flow, and level are important in preventing fires resulting from loss of containment. Instrumentation should be designed to facilitate routine testing. A review should be made of the desired action of each controller, in cases of air or electrical failure, to determine whether the valve should fail open or closed or should remain in its existing position. Guidelines for installation of alarms, recorders, and shutdown systems must consider employee safety and equipment protection. The suitable positioning of such instruments can minimize risk when emergencies arise.

Consideration should be given to the use of an alarm prioritization system, since the number of alarms that may be activated under upset conditions can be high. The need for independent control signals on critical instruments also should be considered.

Plain glass or other types of rotameter and gage glasses that are vulnerable to mechanical or fire damage should be avoided in hydrocarbon or hazardous chemical services.

Guides to the installation and design of process measuring equipment are provided by the Instrument Society of America (ISA)[11]; NFPA *Furnace and Boiler Codes*; and the ASME *Boiler and Pressure Vessel Code.*

3.5 LOCATION

The location of a refinery ideally will provide an area of usable land ample for safe spacing of all facilities, with an allowance for future expansion. Consideration should be given to the nature of adjacent property and its location relative to the refinery, since mishaps at refineries may expose these properties to a variety of potential hazards. Conversely, since adjacent properties can also expose the refinery to different types of hazards, those potential hazards should be taken into consideration when locating the refinery. For example, the location of neighboring airports and the prescribed landing patterns of aircraft could present collision hazards.

[11]The Instrument Society of America, 67 Alexander Drive, Research Triangle Park, North Carolina 27709. www.isa.org

3.5.1 Climate and Geography

Natural perils such as windstorms, floods, and earthquakes can create fire hazards. The frequency and severity of these events should be taken into consideration when designing refinery systems. For example, certain areas subject to potential earthquakes require special bracing of equipment. Areas subject to extreme cold and heavy snow may require special designs to prevent equipment failure due to freezing or excessive snow-loading, along with the resultant fire hazards associated with those conditions.

Prevailing wind direction is generally unreliable. However, facilities with the potential for release of flammable or toxic materials may include an awareness of prevailing wind conditions—to reduce the chance of ignition or exposure—while taking note of the seasonal variance of prevailing wind direction.

3.5.2 Availability of Water

Availability of sufficient fire water and process water from municipal or natural resources are important requirements when considering the location of a refinery. Further details on fire water supply are discussed in Section 4.2.

3.5.3 Supplementary Local Fire Protection

The resources available from local public or private fire departments should be investigated. Well-equipped public fire departments can be of valuable assistance. Generally, these departments do not train for fighting large petroleum fires; so sharing of training facilities and experience can be mutually beneficial. Mutual aid organizations, including other refineries or allied plants located in the same area, should be investigated. Members of such mutual aid organizations frequently have specialized equipment designed for industrial use and have proven beneficial at locations where they have been put into effective operation. The availability of these organizations can influence the amount of private fire protection equipment and staffing required. These frequently function as part of a Local Emergency Planning Committee (LEPC), an organization mandated by SARA Title III; or Community Awareness and Emergency Response (CAER) organization, an initiative of the Chemical Manufacturers Association.

3.6 LAYOUT

The layout of equipment will vary widely, depending on topography, types of units, and equipment to be installed, and, to some extent, the company operating methods. However, in developing the overall layout of a refinery from a fire protection standpoint, it is important to consider the location, spacing, and arrangement of all the various facilities to be installed.

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New facilities should be arranged so that, in the event of a fire or explosion, the potential for personnel exposures is kept to a minimum. In addition, the layout should provide accessibility for firefighting and a degree of area isolation to prevent fire spread within a unit. Consideration should be given to potential impact on neighboring property.

The typical rectangular or block layout provides many advantages. It can provide off-site locations for process piping, service lines, and fire main systems. In addition, future expansions can be completed with a minimum of disruption to existing operations. Roadways that separate the blocks provide excellent fire breaks and facilitate the movement and operation of fire-fighting equipment.

3.6.1 Drainage

When a new refinery layout is planned, preventing the spread of fire must be considered. Suitable drainage must be provided to prevent the spread of major spills from one area to another and to adequately control surface drainage and refinery waste water. The natural slope of the land can often be used effectively in the development of an adequate drainage system. Ideally, storage facilities should be down grade from process areas. Windstorm and flood data must be used, along with potential fire water usage, in determining proper drainage system requirements.

3.6.2 Spacing Of Process Units

The spacing and arrangement of equipment depends upon the type of process unit, the kinds of materials handled, the design philosophy, and the method of operation in a given refinery. In planning the layout, consideration must be given to safety, environmental impact, constructability, economy, operability, and efficiency of process and maintenance operations.

Decisions on spacing between process units should take into account several factors. These can include:

a. Exposure to fire radiation.
b. Exposure to possible explosion overpressure.
c. Access for fire-fighting equipment.
d. Maintenance requirements of one unit while operating another (including activities such as unrestricted hot work).
e. Business interruption potential and the relative value of units involved.
f. Prevailing wind direction.
g. Access for construction and maintenance.
h. Effect on (or from) neighboring installations (community, public roads, heaters, tankage, etc.).
i. Future expansions.

Decisions on distances between flammable and combustible liquids storage and process areas require consideration of many of the same factors as the decisions about distances between process units. Tank dikes provide primary protection. Where possible, consideration should be given to locating process areas on ground higher than tankage, so that in the event of an incident, the topography would prevent major tankage spills, frothovers, or boilovers from gravitating into the process area. See 2.6 for special situation definitions, Section 3.6.7 for further discussion of spacing of equipment outside of process units, and NFPA 30 for minimum separation distances.

Decisions on the separation distances between the areas handling LPG (storage and process areas) and other areas require careful consideration. A release and explosion at one of these storage areas can produce an overpressure causing damage several hundred feet away. See API Std 2510 and NFPA 58 for further spacing information.

3.6.3 Process Plant Buildings

Wherever possible, buildings not directly involved with hydrocarbon processing should be separated from process areas, storage areas, loading facilities and other hydrocarbon handling equipment. For several reasons, including the needs for unit surveillance, maintenance, and process control during emergencies, some process plant buildings are located in or near hydrocarbon processing or handling areas. For information on managing risks associated with these process plant buildings, see API RP 752, *Management of Hazards Associated with Location of Process Plant Buildings.*

3.6.4 Roads

The design of adequate roads for transportation and communication is important in fire protection. Access to all refinery areas should be assured by roads wide enough for emergency vehicle access, positioning, and egress.

Because it may be necessary to block certain roads in emergencies, two or more approaches to each refinery area are desirable. Turning radii should allow adequate space for mobile equipment to clear pipe supports and equipment. Pipe racks and other crossings should be designed to provide adequate overhead clearance for emergency and other vehicles.

Roads through tank fields should be well drained and provided with sufficient turnouts. Slightly elevated roads may be necessary in areas subject to flooding. Pre-fire planning for storage tank areas should consider access for placing fire trucks and high capacity foam monitors.

The location of public or private main roads relative to process equipment should be considered because motor vehicles can be ignition sources.

3.6.5 Egress

For personnel safety reasons, adequate means of exit should be provided from all buildings, process areas, and elevated structures. A review of the need for a second means

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of egress from elevated structures should take into consideration the following:

a. Frequency and number of personnel on the elevated structures.

b. Type, volume, and pressure of the hydrocarbon sources that could restrict egress in the event of a fire.

c. Availability of fire suppression equipment to immediately quench a fire and safeguard the exit routes.

d. Height of the structures and the ability of personnel to move laterally away from the fire hazard.

3.6.6 Location of Equipment Within Process Units

Equipment on a refinery process unit can be arranged in many ways, each having its own advantages and disadvantages. Safety, economy, operability, and ease of maintenance must be considered in locating each item within the unit. The relative importance of these considerations varies with each item, process unit, and refinery, and must be evaluated for each case with equipment located accordingly. Adequate spacing between equipment should minimize the spread of fire. Consideration should be given to access for fire suppression.

3.6.6.1 Process Areas

Areas containing hydrocarbon processing equipment should preferably be located outdoors, allowing open ventilation to dissipate leaked or spilled vapors. Experience has shown that fires and explosions have been prevented or minimized when only a roof and partial wall were used to protect equipment from wind, rain, and snow. Processing equipment should be located as far from ignition sources as possible.

When process equipment is located indoors, suitable ventilation should be provided to prevent the accumulation of vapors in the event of a leak. NFPA 30 provides additional guidance on ventilation design.

3.6.6.2 Pumps and Compressors

Rotating equipment, such as pumps and compressors, is more susceptible to accidental releases than most other equipment. Therefore, from a fire protection standpoint, it is preferable to locate this type of equipment away from ignition sources. Pumps handling hydrocarbons should be carefully located, avoiding areas below pipe alleys, major vessels, air coolers, and other critical equipment. Where this cannot be avoided, consideration of fixed fire protection systems is appropriate (see API RP 2030). Additional spacing should be given to mechanical equipment handling flammable liquids near or above their autoignition temperature, and fixed protection should be considered.

3.6.6.3 Instrumentation

When possible, instrument control cables for critical instrumentation should be routed underground or aboveground through low-risk areas. This will reduce the chance of a minor fire causing an emergency shutdown, and will allow for an orderly shutdown in the event of a major fire. Fireproofing protection should be considered when passing critical cable trays or conduit through areas with a higher potential for fire.

3.6.6.4 Vulnerability to Fire Damage

The vulnerability of certain types of equipment to fire damage is an important layout consideration. For example, air fin coolers represent a greater degree of vulnerability than do tubular exchangers.

3.6.7 Spacing of Equipment Outside of Process Units

3.6.7.1 Atmospheric Tank Storage

Generally, atmospheric tankage is intended only for storage of stocks which, at storage temperature, have a true vapor pressure less than atmospheric pressure. The location and arrangement of tanks will generally be governed by topography, character of nearby structures, types of stock to be stored, shipping facilities, process flow, routing to tankage, and refinery operating conditions.

From a fire protection standpoint, decisions on the layout and spacing of tanks within a tank farm should take into account several factors. These include:

a. Characteristics of the stored product.
b. Size of the tanks.
c. Maximum potential fire radiation.
d. Boilover potential for crude oil and other viscous oil storage.
e. Fixed fire protection provided.
f. Access for fire-fighting equipment.
g. Business interruption consequences.
h. Prevailing wind direction.
i. Distances from adjoining properties where buildings may later be constructed.
j. Future expansions.

Spacing between major tankage and process units should be maximized. For additional information and minimum spacing requirements see NFPA 30.

Hydrocarbon pumps and main control valves should be located outside of the diked area.

3.6.7.2 Pressurized Tank Storage

Major releases or fires impacting aboveground storage of LPG materials in large vessels pressurized above 15 pounds per square inch gauge has the potential for significant

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impact, including BLEVEs, if tankage has fire impingement. Decisions on separation distances between LPG storage and areas such as process units and buildings require careful consideration of the explosion potential of these storage vessels; explosions can cause damage several hundred feet from the storage area. Therefore, it is advisable that these installations be located as far as practical from major process areas. Refer to API Std 2510 for minimum spacing requirements; API 752 for issues dealing with structures; and NFPA 58 for general LPG considerations.

Pumps, piping manifolds, and extraneous aboveground piping should be located outside the dike or spill wall area surrounding the pressure vessels.

3.6.7.3 Pipelines

Areas of potential fire exposure should be avoided when routing main refinery pipelines. Hydrocarbon piping from an offsite facility (such as storage) or from a unit that will remain in operation should not be routed through a unit that can be independently shut down for overall maintenance.

Piping on sleepers should not be exposed at drainage ditches or trenches where oil may exist. Where pipelines cross drainage ditches, flanged or threaded joints should be avoided.

3.6.7.4 Loading Racks

As with other process operations, loading racks should be separated from other refinery equipment so that fires associated with either the rack or the adjacent unit will not spread to the other from their point of origin.

3.6.7.5 Marine Terminal Facilities

Marine terminals are vitally important to the continued operation of some refineries. The location of marine terminal facilities is dependent on the condition of the waterway. However, since these terminals are potentially subject to large petroleum spills, consideration should be given to providing clear space between docks for fire-fighting access. Refer to the OCIMF *International Safety Guide for Oil Tankers and Terminals* for further information on layout and design of marine terminals.

3.6.7.6 Cooling Towers

Cooling towers can be vital to the operation of several process units. Since they are potentially susceptible to fire, some of the same considerations that are used in determining spacing between process areas should be used in determining the location of cooling towers.

When determining the location of cooling towers, wind drift should be considered. This is because water vapor from cooling towers can cause equipment corrosion and poor visibility. It also has the potential for freezing on roads and other areas in cold climate locations. Preferably, cooling towers will be located downwind of process areas, control rooms, and instrument air dryers.

3.7 FIREPROOFING

While location and spacing are of substantial importance in minimizing the degree of equipment involvement in a fire, additional protective measures still may be necessary. One of the key protective measures is the capacity of equipment and its support structure to maintain their integrity during a fire. The purpose of fireproofing is to permit the emergency shutdown of a unit, restrict the addition of fuel to a fire, and protect personnel and equipment from the effects of equipment or support failure during a fire. See API Publication 2218 for details on the design and application of fireproofing in hydrocarbon processing and storage areas.

3.8 PRESSURE RELIEF AND FLARE SYSTEMS

For the design of pressure-relieving systems, refer to API Recommended Practices 520, 521, and Std 2000. For requirements on safety valve installation, set pressures, and valve arrangement, refer to Section VIII, "Unfired Pressure Vessels," of the ASME *Boiler and Pressure Vessel Code*. Subsequent discussions of design will be limited to problems associated with fire and explosion risks involving pressure-relief or flare facilities.

3.8.1 Relief Valves

Safety-relief valves are provided to prevent overpressure of refinery equipment in emergencies created by operational errors, equipment failures, and fire; and to ensure the safe disposal of released materials. Normally, relief valves that release liquid hydrocarbons should discharge into a closed disposal system. Where only hydrocarbon vapor is vented, relief valves may discharge directly to the atmosphere if certain conditions are met. These include the following:

a. Low toxicity.
b. Temperature below autoignition.
c. Dispersion of released material based on discharge velocity, molecular weight, condensing temperature, and other physical properties of the material.
d. Proper distances between points of release and personnel areas.
e. Relevant environmental regulations for emergency releases.

Safety relief valves should be tested at scheduled intervals to determine whether or not they will function at the designated pressure. The test interval will vary from several months to several years, depending upon the cleanliness of the service, the severity of operating conditions, the results of

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previous inspections, and the existence of legal requirements. The frequency of the test should be established for each valve on the basis of operating experience, engineering judgment, and industry practice (see API RP 576).

Where relief valve discharges are routed to a closed system, headers should be sized to handle the maximum flow resulting from any single contingency without developing excessive back pressures on the relief valves connected to the system. Where safety relief valves are connected into closed emergency relief systems, the grouping of equipment within one fire risk area may influence the size of the relief facility piping required. By providing reasonable separations between groups of equipment, the quantity of materials released into the flare systems under fire conditions can be reduced. Fire-related electric power failure should be included in flare system contingency analysis.

Conditions under which the system can become plugged, blocked, or frozen should be avoided; and piping should be designed without "pockets" and sloped toward liquid knock-out drums to avoid the accumulation of liquid that may cause surging, plugging, or freezing.

If a closed emergency relief system serves more than one unit, it may be necessary to isolate sections of the system when individual units are shut down. Therefore, the location of adequate facilities for blinding or isolating the unit should be examined.

When safety-relief valves discharge to the atmosphere, vent stacks should terminate above adjacent equipment. Otherwise, localized overheating of nearby process vessels might result from radiation or convective heating if a valve should open, and vapors ignite, during a fire or other abnormal condition.

Drainage on the discharge risers of atmospheric safety valves should be designed to reduce the likelihood of problems. If drains become clogged, condensed hydrocarbons from leaking safety valves can accumulate and subsequently be blown over the unit if the valves release. If the material reaches an ignition source, a fire could result. If water accumulates in the vent stack, freezing may occur, either as a result of low ambient temperatures or from the refrigeration process caused by rapid evaporation of leaking low-boiling-point hydrocarbons.

Drains should be installed at the bottom of the riser. Care should be taken with regard to drain orientation so that, under fire conditions, any burning vapor discharged from the valve opening will not impinge on equipment. Installation of elbows in the drain to direct any released vapor away from nearby vessels or piping can reduce this possibility.

Vent stacks that discharge at elevated locations are sometimes equipped with snuffing steam to extinguish fires caused by lightning-ignition of relief valve leakage. See API RP 520, Part II, for additional information.

3.8.2 Flare Systems

Flare systems provided for the safe disposal of gaseous refinery wastes are the subject of API RP 521. Depending on local environmental constraints, these systems can be used for the following:

a. Venting during startup or shutdown.

b. Venting of excess refinery gas.

c. Handling emergency releases from safety valves, blowdown, and depressuring systems.

Designs vary considerably, depending upon the type of connected equipment and the complexity of the overall system. Though specific design details are not discussed in this publication, some potential problem areas are reviewed.

Because flare systems are open to the atmosphere and a continuous ignition source exists at the end of the line, internal system explosions are possible if appropriate safeguards are not in place. The most positive approach to eliminating this hazard is to adequately purge the system to remove trapped air before the pilots are ignited. Provisions should be made for the safe lightoff of these pilots, either by electrical ignitors or by ignition pipes, where flammable mixtures can propagate the flame to the pilots.

Vent connections or bleeders that could allow air to be released into the flare system should be avoided. However, the total elimination of air from flare systems may not always be practical. To minimize the risk, it is common practice to either continually inject a fuel-gas purge or to install liquid seals as flashback devices to limit flame propagation through the system. For this purpose, liquid seals are preferable—and the use of flame arrestors is discouraged—because the small passages in flame arrestors are subject to fouling that may cause excessive pressure drops during periods of high flow. The design of purging systems should consider the possibility of air ingress into the flare system from the flare stack when using lighter-than-air purge gases. Dry molecular seals can be used to minimize purge gas rates.

Another area of concern is the possibility of liquid carryover from the flare stacks that may spray burning liquid over a widespread area and start fires at grade level. The size of knock-out drums must be based upon the equipment that discharges to the drum, the rates of liquid entry, the duration of a release, and whether or not blowdown or liquid pump-out streams are routed to the flare stacks.

Consideration should be given to the provision of a high-level alarm and liquid pumpout on the flare knock-out drum. Maximum allowable liquid levels must be established to ensure adequate vapor space to avoid entrainment of liquid when safety valves operate. System safety reviews should consider the effect on downstream storage of potentially high-temperature effluent from the knock-out drum.

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The location and height of flare stacks should be based upon the heat-release potential of a flare, the possibility of personnel exposure during flaring, and the exposure of surrounding plant equipment. Where personnel are expected to work in the vicinity of a flare without restriction on the length of exposure, 500 BTU/hr-ft^2 (1.58 kW/m^2) radiant heat exposure level has been established by some companies as an acceptable limit. However, radiant heat levels up to 2500 BTU/hr-ft^2 (7.9 kW/m^2) may be tolerated for 5 to 15 seconds if the only concern is short-time exposure of personnel to permit escape from the area under emergency release conditions.

The location of flares in the vicinity of tall refinery equipment should be examined. Flames or hot combustion products can be carried by the wind, which could cause problems and create hazards to personnel working on these elevated structures at the time of a flare release.

Another important consideration in the design of flare systems is the adequacy of drum seals that might have open sewer connections. Liquid levels are frequently maintained in drums by a loop-style seal. The highest pressure that may exist in these drums under maximum flare release conditions should be determined. The seal leg should be of adequate depth to prevent displacement of seals under the highest pressure in order to avoid release of hydrocarbons at grade level or to the sewer.

3.9 DRAINAGE, CONTAINMENT, AND WASTE DISPOSAL

The design of drainage and the determination of the means and capacity required to control refinery wastes are of prime importance in minimizing the size of liquid hydrocarbon fires. Maximum fire water runoff, rain runoff, and spill size requirements should be considered in designing drainage and containment systems.

3.9.1 Spill and Waste Disposal for Process Areas

From a fire protection standpoint, the purpose of drainage and containment systems in process areas is to minimize the area subject to hydrocarbon spills and to direct those spills away from critical equipment. Process areas are generally paved with a hard surfacing material. The pavement should slope toward catch basins located in open areas so that spills will flow away from the equipment rather than under it. All spill and waste disposal systems must consider social and regulatory environmental constraints during their design.

Fire stops or water seals should be provided throughout industrial sewer or drainage ditch systems to prevent the vapors from material released into the sewer system from spreading to other areas. Sealed sections of sewer systems should be vented at suitable points to prevent pressure buildup and the blowing of seals if light materials should enter the system. The sealing and venting of such systems is of particular importance where drains from a building are connected to an industrial sewer system.

Below-grade trenches, compressor or pump pits, and building basements within process areas should be avoided where practical. These areas present corrosion and housekeeping problems and provide a place for trapping flammable vapors.

Open refinery separators are generally located away from process and storage equipment. However, this may not always be feasible, so care should be taken to minimize locating ignition sources near these open separators because, under some conditions, flammable vapors could be released from their surfaces.

Discharges to the sewer from process equipment handling volatile hydrocarbons can be a mechanism for vapor release. If a level controller were to fail in an open position it could be a source of vapor in the immediate process area, as well as throughout the sewer system.

As noted above, vapor accumulation can pressurize sewers, resulting in blown seals and vapor release at other points. Where such potential exists, suitable instrumentation or a disengaging drum should be provided.

3.9.2 Diking and Waste Disposal for Storage Areas

For low-flash-point refined oil and crude oil storage tanks, dikes are usually provided to prevent oil spills from involving other facilities. The usable volumetric capacity of the diked area should not be less than the greatest amount of liquid that can be released from the largest tank within the diked area, assuming a full tank. See NFPA 30 for requirements of dike design and arrangement.

Dikes are used primarily to contain tank overflows, bottom leaks, and line failures, as well as to limit the uncontrolled off-site flow of water applied during a fire. Accumulations of liquid will either be limited in quantity or be removable before the retention capacity is reached. Experience has shown that, under normal conditions, it is unlikely that the full dike capacity would be needed.

Should a total tank failure occur, the oil wave may not be contained, and oil could surge over the dikes. This could also be true where boilover or slopover conditions prevail.

To permit ready movement of fire-fighting equipment across the dike, most companies restrict dike heights to about 6 feet (1.8 m), although higher dikes are permissible.

In the event of a fire, it is advisable to develop a means of controlling drainage from the dike area. Pipe connections through the dikes, valved on the exterior of the dike and discharging into drainage ditches, should be an adequate means of control.

Multiple tanks within one diked enclosure are acceptable. However, intermediate dikes (curbs) may be necessary to separate tanks or groups of tanks. NFPA 30 provides additional details and presents impounding alternatives to diking.

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3.10 POWER AND UTILITIES

Refinery service and utility lines, such as water, steam, instrument, air, electric light, and power, should be designed for maximum reliability. A reliable communications system also is necessary. An alternate communications system for use during an emergency may be desirable. The possibility for service interruption of modern computerized digital telephone systems, which rely on electric line power, should be considered during utility reliability studies.

3.10.1 Electric Power

API Recommended Practice 500, NFPA 30, NFPA 497A, and NFPA 497B should be used as guides in determining the proper electrical area classification. Local electrical codes or the *National Electric Code* should be used for design and installation within the specific classified areas. Small units can use NFPA 497A where only small amounts of hydrocarbons are present.

Power lines may be installed overhead or underground, depending on applicable regulations and local conditions at the site. In a relatively unobstructed area such as a tank farm, a pole or tower system may be the most practical. An underground system using conduit or trench-laid armored cable may be preferable in congested areas.

Pole line and underground systems should be located, where possible, along roads or pipelines to avoid the need for relocation in the event of future expansion.

The use of outdoor substations in nonhazardous areas reduces the possibility of flammable vapor concentrations that might occur in substation buildings from ground seepage and poorly sealed drains.

In the event of fire, circuits may be subject to interference from exposure to high temperature or water drainage. This should be considered when locating overhead lines or when establishing a safe location for a substation in process areas.

Electric power may be purchased from a utility company or generated in the refinery. The choice depends upon the availability and cost of reliable power purchased from a utility company. If power is purchased, it is preferable to have at least two main feeders to bring power into the refinery via two different routes from the utility company's generating plant. Each feeder would normally carry half the plant load, but each would have the capability to carry the full load. Automatic switchover, with motor controls arranged so that a momentary power loss would not result in motor shutdowns, could then be provided in the event of the loss of one feeder. If power is generated in the refinery, the generators may be steam, gas turbine, or diesel-driven. Where the possibility of electric power outages exists, small engine-driven emergency generators may be provided for critical electrical circuits. Emergency equipment may either be manually operated or arranged for automatic switchover upon failure of the main supply. Dependable AC power for critical electrical control circuits may be provided by battery sets equipped with chargers and static inverters.

3.10.2 Steam

If steam is used to generate electrical power, steam-generating facilities should be located where they will have minimal exposure to external fire. The facilities should be designed to allow automatic shutdown of noncritical steam users following the loss of electric power, thereby reducing the steam load during power failure.

Steam boilers may be weather-protected and installed outdoors. When the boilers are housed, the building should be constructed of fire-resistant or noncombustible material. Area drainage should be designed to prevent an inflow of flammable hydrocarbons.

The boiler fuel system should be designed to reduce the possibility of explosion in the event of burner flame failure. The modern, high-heat-release, fast-steaming boilers that use a forced draft do not allow sufficient time for the operator to determine the corrective action required in an emergency. Therefore, fully automatic boiler supervisory controls with continuous burner flame scanning should be considered.

It is preferable that main steam lines be installed in areas of low fire exposure, such as along roadways. A looped system designed with division valves to permit shutoff of sections and to permit steam to be supplied from two directions is desirable. Steam lines are usually installed overhead to simplify condensate drainage and insulation requirements and to provide for pipeline expansion. It is advisable to fireproof the overhead supports, or to provide water cooling where they are located, in areas subject to severe fire exposure.

3.10.3 Water

Water pumps for process, fire, and service requirements should be located as close as possible to the water supply source. This will permit the use of short suction mains for effective and reliable pump operation. Steam, motor, or internal combustion engine pump drives may be used, depending upon the reliability of the power source. Multiple pumps, each using a different type of drive, often provide optimum reliability. Total capacity should be sized based on normal needs and emergency scenario analyses.

There are two important considerations: continuous supplies of boiler feed water for steam; and cooling water to prevent excessive vaporization in process equipment. In an emergency, depending upon the plant design, these needs may be equally as important as water for fire suppression.

Spare engine-driven water pumps are desirable where the possibility of electric power outages exists and generation of refinery steam to operate water pumps can be subject to interruption.

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Loss of makeup water to a cooling tower could eventually result in a sudden loss of cooling water to the process units being served. Since it is possible that a makeup water level control valve could stick in the closed position, some refineries install low-level alarms on the cooling tower basin.

Hydrocarbon vapor release is another consideration with process water cooling towers. This may be caused by hydrocarbon leakage from coolers and condensers into the cooling water return system. Provision for vapor disengaging and venting from the return system may be considered. Detection of hydrocarbon leakage, location of cooling towers, and adequacy of vapor dispersion should all be considered in the design of process water cooling towers.

Measures should be taken to prevent freezing of water mains that provide the necessary fire, process, and service requirements. Interconnecting or looping of lines in vital services is desirable, but process and fire water systems should be separate whenever possible. If the systems are connected, precautions should be taken to prevent the loss of fire water. Contamination of this fire water by process water that may contain hydrocarbons, or additives that might cause foam breakdown, should also be prevented. Refinery fire water systems are discussed in Section 4.2.

4 Fire Control and Extinguishing Equipment

4.1 GENERAL

This section will discuss the general approach for fire control and extinguishment that has been successfully adopted for use by both large and small facilities throughout the petroleum industry. The goal of effective fire control is to extinguish a fire in the shortest possible time, with no loss of life, and minimum loss of property. The exception to aggressively seeking rapid extinguishment is where the fuel for the fire is a pressurized gas or liquid release which should be allowed to burn until the source of fuel can be shut off.

The primary objective of firefighting is to extinguish a small fire before it expands to become a large fire; or to control a large fire and protect adjacent exposures until emergency response staffing is sufficient to safely mount an aggressive suppression effort.

Most facilities have three basic types of fire-fighting equipment ready for immediate use:

a. Fixed System: A fire protection system that is permanently installed and connected to a supply of extinguishing agent(s). These systems may be automatically or manually activated. An example of a fixed system would be a water spray system that is supplied directly by the plant fire water system.

b. Semi-Fixed System: A fire protection system that is permanently installed but not connected to a supply of extinguishing agent. These systems generally require personnel to manually connect an extinguishing agent supply to the system prior to use. An example is a tank foam system that terminates at a connection located at the dike wall.

c. Portable Equipment: Fire suppression equipment that must be moved to the site of the fire, manually assembled, or positioned before being put into service. It is generally stored until needed at a location accessible to its intended users. Examples include fire trucks, fire hose, foam monitors, fire extinguishers, and most fire department equipment.

Fire protection equipment should be kept in fully functional condition and tested periodically in accordance with accepted procedures. In some instances, equipment may be regularly inspected by an outside agency. Proper records should be kept of each inspection (see NFPA 25).

4.2 WATER FOR FIRE SUPPRESSION

Water is used universally as a fire-fighting agent. It serves as a cooling, quenching, smothering, emulsifying, diluting, and displacing agent. The high latent heat of vaporization of water (its high absorption of heat when converted to steam) makes it particularly valuable in fighting oil, gas, and Class A fires. It is usually available, easily handled, and when applied as a fog (finely divided spray), is effective and safe to use on most petroleum fires—where it inhibits combustion by both cooling and smothering (excluding oxygen).

Water is the primary agent for cooling equipment, structures, and tank shells that are exposed to a fire. This prevents or reduces both heat damage to equipment and overpressure that results from overheating vessel contents.

When used as a coarse high-velocity spray stream, water can sweep pools of burning fluid out from under elevated equipment. Extinguishment will result if the fuel surface can be cooled below the temperature at which it will give off sufficient vapor to support combustion (fire point). The proper way of applying water for extinguishment is in the form of a spray. If a water spray is properly applied to the surface of a burning liquid hydrocarbon with a flash point above 200°F (93°C), it can produce a layer of froth on the liquid surface, which will act similar to foam and smother the fire. Fighting a hot asphalt tank fire is an example. However, care must be used to prevent slopover.

Spills of flammable liquid that are soluble in water may, in some instances, be extinguished by dilution. This same approach may be impossible on fires within tanks because overfilling may occur before sufficient dilution is achieved. The quantity of water required to extinguish a fire by dilution varies with the solubility of the products and is generally quite large. For example, a solution of 75 percent water and 25 percent ethyl alcohol will support combustion when hot. As a result, for soluble materials it is generally impractical to extinguish fires involving deep liquid spills, and those within

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tanks, by dilution with water because fire spread and tank overfilling may occur before sufficient dilution is achieved.

Water is used as a displacement medium in leaking hydrocarbon lines. It may also be used to float liquid hydrocarbons above a leak in a tank to replace product leakage with water leakage. To be effective in pressurized pipes and tanks, the water pressure must be greater than product pressure. Care should be taken to avoid flow of a higher pressure product into a lower pressure water system and to avoid overpressure of the vessel and piping. Water cannot be used to displace refrigerated liquefied petroleum gas or liquefied natural gas from a leaking pipeline if the product temperature is colder than 32°F (0°C), or for liquids that have a temperature above 200°F (93°C).

By cooling exposures and controlling fire intensity, water can be used effectively to control pressurized gas or liquid fires and spill fires involving low-flash-point fuels. Water cannot be used effectively to extinguish such fires and extinguishment may be undesirable because of potential vapor cloud hazards. However, experienced fire-fighting personnel can use water spray as very effective personal protection from radiant heat and flame contact to gain access to equipment so that valves can be closed, shutting off the fuel source for fire suppression. And, in some conditions, firefighters can disperse moderate quantities of escaping gas or vapor using water spray.

Water is the principal ingredient in fire-fighting foam; mixed with foam concentrate, this is the most effective agent for extinguishing large flammable liquid spill fires or tank fires.

4.2.1 Water Supply

The required water supply for firefighting may be obtained from a combination of dedicated higher-pressure fire water mains, lower-pressure processing or cooling water mains, and other available and reliable sources. Each important fire risk area should be looped with fire mains sized to supply the critical areas with water at the fire flow rate and pressure.

The fire water system should be designed such that the fire main sizing meets fire flow requirements throughout the refinery. The demand should be based on hydraulic calculations for current and future demands, with a minimum fire main size calculated to deliver a minimum of 75 to 125 psig (520 to 940 kPa) hydrant pressure at full-rated demand.

Gate valves should be installed to sectionalize the water main grid so that only part of the system will be out of service during failures or repairs. Gate valves and buried system post indicator valves should be used and marked for easy identification.

Fire water should be supplied by a system that is independent of all other uses and be from a reliable source. If the normal source is not reliable, an emergency supply should be provided from a storage tank or reservoir. If a storage tank or reservoir is used for emergency water, it should normally be kept full. The storage tank capacity should be adequate to meet the high-pressure water requirements for a period of at least four to six hours. It is desirable to have an adequate supply of makeup water available in addition to storage requirements.

Water sources should be of high quality and free of water treatment chemicals that will be detrimental to fire-fighting foam generation. Pump inlets must be adequately protected from debris entering the system. The use of all possible water sources for firefighting may provide large economies in construction and operating costs. Sources for consideration are: low-pressure process mains, the public water supply, and natural water sources. Also, other available sources of water, such as storm water drainage sewers and environmental ponds, should be considered. Under emergency conditions the use of cooling water tower basins can be considered. Regardless of the source, the water supply should equal the minimum requirement under adverse conditions.

Fire lines, valves, hydrants, monitors, pumps, storage tanks, and portable water-type extinguishers should be protected from freezing. Lines should be designed and installed to avoid overstressing caused by earthquakes, settlement, severe shock from mechanical impact, and damage by fire exposure. Diagrams of lines and valves should be posted at key points throughout the plant, and any change or alteration to the system should be recorded concurrent with the change.

Connections to fire-fighting systems that permit the diversion of water for other purposes should be avoided. Where such connections are in place, procedures should be established to ensure that adequate volumes of fire water should be available when needed.

For more information, refer to NFPA 24, *Installation of Private Fire Service Mains and Their Appurtenances.*

4.2.1.1 Flow Rate Ranges for Manual Firefighting

The minimum fire protection water supply should be capable of providing the flow rates, pressures, and duration determined from a pre-incident scenario analysis. The following discussion provides broad general guidance.

The flow rate of water required for fire protection should be calculated separately for each considered fire area within the refinery. The fire protection water supply is normally sized to be capable of providing the largest calculated flow rate required for any single fire area within the facility. Specific flow rates depend on plant design, configuration, and process hazards. The actual design can be determined by 1) providing 0.1 to 0.5 gpm/ft^2 (4.1 to 20.4 lpm/m^2) of water to the fire area based on congestion and unit structures while considering the appropriate fire response based on pre-inci-

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Table 1—Example Water Flow Rates for Manual Fire Fighting[a]

Scenario Area of Interest	Firewater Flow Ranges, Per Minute in Thousands of Gallons (Thousands of Liters)	Example Flow Rate Ranges Based on Protected Area GPM/Ft^2 or LPM/M^2
Radiation heat protection		0.1 gpm/ft^2 (4.1 lpm/m^2)
Process areas handling flammable liquids or high pressure flammable gases	4,000 to 10,000 gpm (15,000 to 38,000 lpm)	Cooling: 0.2 to 0.3 gpm/ft^2 (8.2 to 12.3 lpm/m^2) Suppression: 0.3 to 0.5 gpm/ft^2(12.3 to 20.4 lpm/m^2)
Process areas handling gases or combustible liquids	3,000 to 5,000 gpm (11,000 to 19,000 lpm)	0.20 to 0.30 gpm/ft^2 (8.2 to 12.3 lpm/m^2)
Tank storage of flammable and combustible liquids in atmospheric tanks		See API 2021 & NFPA 11
LPG storage tanks and vessels		See API 2510, API 2510A and NFPA 58
Warehouses		See applicable NFPA Fire Codes
Buildings, offices, laboratories, and similar structures		See applicable NFPA Fire Codes

Note: [a]The total (gpm or lpm) flow required will depend on size, congestion and the needs of the exposed facilities being protected. The specific flow rate (gpm/ft^2 or lpm/m^2) chosen will depend on the definition of the fire area and the fuel loading in the area.

Table 2—Suggested Residual Pressures

System or Equipment Item	Desired Residual Pressure	Measured At:
Calculated sprinkler, water spray or foam system	As determined in the system calculations, generally between 75 and 125 psi (520 to 940 kPa)	As specified in the calculations, usually at the base of the system riser or the connection to the main.
Fixed monitor	100 psi (690 kPa)	At the monitor base.
Hydrant directly feeding hose streams	100 psi (690 kPa)	At the hydrant.
Hydrant feeding a fire truck if no fixed equipment is operating	20 psi (138 kPa)	At the hydrant

dent plans and experience, or 2) historical experience with similar facilities.

Where the fire protection water system is intended to supply only monitors and hose streams in support of manual fire fighting and suppression, sample flow rate ranges may be estimated using values suggested in Table 1.

4.2.1.2 Determining Water Flow Rates for Process Areas

Where the considered fire area is totally or partially protected by fixed water spray, sprinkler, or foam systems, the flow rate should be the sum of the flow rates required for proper operation of the fixed systems, plus an allowance for simultaneous operation of monitors and hose streams. Where there are multiple fixed systems within the fire area, the calculated flow rate should consider whether adjacent systems may need to operate concurrently.

4.2.1.3 Suggested Residual Pressure

The residual pressure required for fire protection should be determined separately for each considered fire area within the refinery. The required pressure should be the highest pressure required by any system or piece of equipment at the delivery point where it would be operated during the fire scenario. Suggested residual pressures for common fire protection systems and equipment items are shown in Table 2.

The fire protection water supply and the distribution piping should be sized such that the required pressure is met at each considered fire area when flowing the required flow rate for that fire area.

4.2.1.4 Fire Water Flow Duration

The total water supply within the refinery should be capable of supplying the maximum flow for a period of not less than four to six hours, consistent with projected fire scenario needs. Where the water system is supplied from a tank or reservoirs, the quantity of water required for fire protection should be reserved exclusively for fire protection. However, where the tank or reservoir is automatically filled by a line from a reliable, separate supply, such as from a public water system or wells, the total quantity in storage may be reduced by the incoming fill rate.

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4.2.2 Pumping Equipment

It is desirable to provide firewater pumps with different power sources for drivers to decrease the vulnerability of the system. Most locations divide drivers between electric motors and diesel engines. Steam turbine drivers are also used. Gasoline engine drivers are not normally used for fixed installations because of relatively high maintenance and lower reliability. Centrifugal pumps, including horizontal split case and vertical turbine pumps, preferably with a relatively flat characteristic curve, are generally used for supplying fire water systems. Centrifugal pumps with these characteristics provide a steady, non-pulsating flow of water at uniform pressure and can also idle against closed valves for a period of time without damage to the pump or connected equipment. In some instances, relief valves or governors are provided to prevent over-pressuring the fire water system. Ideally pumps are located in areas not likely to be impacted by a fire or explosion. Larger installations with multiple pumps may place pumps in separate areas of the facility to reduce their vulnerability.

New fire pump installations should meet the requirements of NFPA 20.

Fire system pumps are usually divided into three categories:

a. Jockey pumps that maintain sufficient pressure and have the capacity to supply first-aid hose streams that might be required during emergencies or during the early stages of a fire.

b. Main fire pumps that are operated, either manually or automatically, to provide the basic capacity.

c. Pumps in standby service to provide water in the event of failure or demand exceeding the capacity of other pumps.

4.2.3 Fire Hydrants and Reels

The type of fire hydrant used will depend upon climatic conditions. Standard self-draining fire hydrants meeting the American Water Works Association (AWWA)[12] standards are generally used in freezing climates.

Fire water supply manifolds can be used instead of fire hydrants in non-freezing climates. Fire hydrants or headers can include a variety of outlets, including large "steamer" or hose connections. Hose connections should be compatible with those used on the hoses of the local fire department and mutual aid organization, or adequate adapters should be provided.

The number of hydrants in an area is usually determined by the design and type of process. The normal distance between fire hydrants ranges from 150 to 300 feet (45 to 90 meters), depending on the fire hazard, layout of the area, water requirements, and the number of outlets of the hydrants. The distance between a fire hydrant and a building or structure that is to be protected should be at least 50 feet (15 meters), if possible.

[12]American Water Works Association Headquarters, 6666 West Quincy Avenue, Denver, Colorado 80235. www.awwa.org

4.2.4 Fire Hose

Fire hose may be categorized as attack hose or supply hose. Traditionally most refineries have utilized 2½-inch (6.35 cm), 1¾-inch (4.45 cm), and 1½-inch (3.81 cm) hose as their standard. Large diameter hoses 4 inches (10.16 cm) and 5 inches (12.7 cm) in diameter) have growing acceptance and application. Coordination of hose fittings and connections with local mutual aid or public resources is appropriate irrespective of the sizes of hose used.

The 2½-inch (6. 35 cm) hose is used primarily for heavy cooling streams. Three or more people are required to handle 2½-inch (6.35 cm) hose lines, depending upon the system pressure. 2½-inch (6.35 cm) hose lines are suitable to supply water for portable ground monitors. It is general practice to carry this hose on mobile equipment to reduce plant hose requirements. Some 1¾-inch (4.45 cm) or 1½-inch (3.81 cm) hose is usually carried on the mobile equipment for use in tank farms or other outlying areas.

For rapid availability of water in process areas, permanently connected hose reels and hose stations have been used extensively. These reels are usually provided with a 1½-inch (3.81 cm) firm hose or 1½-inch (3.81 cm) fabric hose for incipient firefighting. The firm hose has the advantage that it can be charged and used by one person without unreeling it entirely. However, its use is limited because hydraulic pressure loss limits the length of hose to approximately 100 feet (30 meters) to assure adequate nozzle pressures.

In some cases, a large diameter hose (4 inches (10.16 cm) and greater) provides advantages:

a. Its low friction loss reduces the number of pumping apparatuses required.

b. It minimizes manpower requirements by reducing the need for dual or triple hose lays.

c. It facilitates the use of high flow rate monitors used for tank fire fighting.

4.2.5 Nozzles

Combination fire hose nozzles providing water patterns continuously variable from straight streams to water spray (or water-fog) are widely used. Straight water streams are used primarily for cooling equipment, although they have been used effectively for sweeping burning hydrocarbon spills away from exposed equipment. Straight water streams are reasonably good at nozzle pressures as low as 45 to 50 psig (310 to 345 kPa), and provide a longer range than water spray nozzles.

Straight stream patterns are generally used when:

a. Extended reach is required, particularly in high wind situations.

b. Deep penetration of the fire area is required.

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c. Water quality is such that fog nozzles cannot be operated due to plugging.

d. Low water pressure prohibits the use of fog nozzles.

It is general practice to use shutoff-type nozzles so that the hose line may be controlled by the person at the nozzle. Most water fog nozzle manufacturers recommend nozzle pressures at 100 psig (690 kPa) with functional operation in a range from 70 psig (485 kPa) minimum to 125+ psig (860+ kPa). When used by experienced personnel, water spray can be effective for:

a. Extinguishing fires involving high-flash products.

b. Chilling hot vapors and preventing ignition.

c. Cooling low-flash-point (gasoline and liquefied petroleum gas) oil and gas fires to reduce their intensity.

d. Cooling exposed equipment and preventing fire damage or involvement of other fuels as a result of blown gaskets and the like.

e. Dispersing vapors and preventing their ignition from nearby ignition sources.

f. Cooling fires in light hydrocarbons service (such as gasoline) so that they may be more readily extinguished by dry chemical agents used in combination with the water supply.

g. Frothing out fires involving heavy viscous oils.

h. Providing a protective cooling curtain for personnel who are operating valves or other shutoff devices.

Reference should be made to manufacturers' literature, NFPA *1964 Spray Nozzles (Shutoff and Tip)* and *Fire Protection Handbook*, for a thorough discussion of nozzle types and water spray patterns, as well as water capacities, range, and related information.

4.2.6 Monitors and Master Stream Delivery Devices

Monitors may be equipped with various delivery devices. They can be mounted on trailers or pickup trucks to provide high volume mobile master streams. Fixed monitors may be desirable for use at spot locations. Because their area of coverage is limited, fixed monitors should be carefully placed to provide optimum coverage. These locations should be reviewed during both the design and the final construction stages of a job to avoid unnecessary obstructions like piping and support columns. In some cases, fixed monitors are provided with shields to protect the operator from radiant heat. Fixed monitors can be configured to provide foam for rapid response to leaks or spills.

The installation of fixed monitors in an area does not necessarily eliminate the need for portable monitors and hydrants for hand line use, because fixed monitors may not provide complete coverage of the entire area and may possibly be inaccessible during an emergency. Portable monitors can be used in place of hand-held lines to reduce stress on firefighters if mobility is not required.

Special installations may require elevated monitors, remote control monitors, or oscillating monitors. Elevated monitors can be used to protect areas inaccessible from grade because of congestion, or at docks.

4.2.7 Water Spray Systems

Water spray cooling systems can be used to reduce fire exposure damage by keeping a water film on exposed surfaces to absorb radiant heat. They can also be designed to reduce fire intensity at the source. Reference should be made to API RP 2030 and NFPA 15.

4.3 FOAM

Foams for fire protection purposes are an aggregate of air-filled bubbles that will float on the surface of a flammable liquid. They are made from aerated solutions of water and a proper proportion of foam concentrate. Foams form a cohesive floating blanket on the liquid surface that extinguishes the fire by smothering and cooling the fuel. They also prevent reignition by inhibiting vapor release and thus preventing the formation of combustible mixtures of vapor and air. Applying a foam blanket to fuel spills before ignition can prevent spills from becoming fires.

Foams are particularly suited for extinguishing two-dimensional flammable liquid spill fires or in storage tanks where the foam forms a vapor-sealing blanket that secures the area after extinguishment. In fires involving jetting or falling fuel, such as an overflowing tank or line flange leak, foam is effective only on the fire that arises from fuel spills and on pools that form flat surfaces. Foam is not suitable for extinguishing fires which involve flammable gases or liquids containing large amounts of liquefied petroleum gas.

To extinguish deep tank fires or spill fires, continuous foam application at no less than the required rate is critical. See NFPA 11 for required rates. Unless the required rate of foam is applied and maintained on the liquid surface until it is sealed by a cohesive blanket of foam, the fire may not be extinguished. Cooling water applied to a tank shell can assist in extinguishing a fire. When foam is applied by hand lines or monitors over the rim of a tank, wind and thermal updrafts frequently carry a portion of it away. As specified in NFPA 11, foam must be applied at a 60 percent higher rate to compensate for this loss.

Cases have been reported where the application of foam through solid streams that plunged into unignited flammable liquids have been thought to be the source of ignition (of an ensuing fire). The ignition has been attributed to static discharges resulting from the splashing and turbulence. Therefore, any application of foam to an unignited flammable liquid should be gentle. Proper application methods with portable equipment could involve use of a spray pattern or banking the foam stream off a backboard or structure so that the foam flows gently onto the surface of the liquids. Prop-

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erly-designed fixed foam chambers on tanks have not experienced problems with static electricity generation.

Foam is made by mechanical action or agitation achieved by the turbulent mixing of atmospheric air into the foam solution of water containing liquid foam concentrate. A variety of foam types typically are used in a refinery:

a. Protein.
b. Fluoroprotein.
c. Special purpose.
d. Alcohol resistant.
e. Aqueous film-forming foams (AFFF).

Factors influencing the choice of a specific foam type for use at a facility include the following:

a. The type of product for which fire suppression may be required.
b. The type of foam concentrate stocked by potential mutual aid participants.
c. Sources and timing of supply from manufacturers for major incidents.
d. Cost effectiveness of the candidate foam concentrates.

The manufacturer of the foam-making equipment should be consulted for the correct percentage of concentrate to be used in their particular system. Proportioners should be designed and set for the percentage of foam concentrate used to form the desired solution.

For detailed information on foam application methods and equipment, see technical information provided by the foam concentrate and apparatus suppliers.

For detailed information on foam types see API Publication 2021 or NFPA 11.

4.4 DRY CHEMICALS

Application of a dry chemical can be effective in controlling and extinguishing fires occurring during the processing and handling of flammable liquids and solids. The finely divided chemical produces free radical interceptors which break the oxidation chain reaction, thus inhibiting the combustion process within the flame itself.

These agents are effective on small spill fires and on fires involving jetting or falling fuels. However, caution should be exercised when extinguishing pressure fires to ensure that the remaining hazard is not greater than the original fire. Dry chemicals are nonconductive agents suitable for fires involving energized electrical equipment with a recognition that the particulate residue may be corrosive and certainly requires cleanup.

Rapid fire control and flame reduction may also be achieved by using multipurpose (ABC) dry chemical on combustible materials such as wood and paper; however, an additional quenching agent such as water must often be used to extinguish the remaining embers.

Dry chemical extinguishing agents have proven to be effective when used simultaneously with water fog without interfering with the effectiveness of the chemical. The water fog will quench embers, cool hot surfaces and reduce flame size making the fire easier to extinguish with the dry chemical. This cooling effect is particularly important if a fire has been burning for a significant period, since there is a high chance of reflash if fuels contact heated metal. For further information, see API Publication 2021, NFPA 10 and 17.

4.4.1 Dry Chemical Extinguishers and Equipment

Dry chemical extinguishers are available in hand-carried, wheeled, truck-mounted, and stationary units. They have capacities ranging from 1.5 pounds (0.7 kg) to 3,000 pounds (1365 kg) of chemical per unit. Multiple units can provide higher capacities if required for special installations. Stationary units can be piped for manual, semiautomatic, or automatic control. NFPA 17 provides information concerning installation of dry chemical extinguishing systems.

Portable hand extinguishers containing 30 pounds (13.65 kg) or less of dry chemical are recommended for use as incipient fire fighting equipment for small fires. Several hand extinguishers may be used simultaneously for extinguishing larger fires. Reserve, or secondary, protection can be provided by wheeled or stationary extinguishers with capacities up to 350 pounds (160 kg). Some operators prefer to use several hand units simultaneously from different angles to provide coverage of the fire and reduce the potential for reignition. This multiple application may be supplemented by properly directed, waterfog streams.

Portable extinguishers should be placed in locations which are safely accessible in the event of a fire. Whenever an extinguisher is used, it should be replaced and removed for inspection and recharging. Some operators seal an extinguisher or cabinet so that it may be monitored more readily. Other operators place the extinguisher in an expendable plastic bag that serves the same purpose while keeping the extinguisher clean and preventing atmospheric corrosion. Reference should be made to the manufacturer's recommendations for inspection, servicing, and repair; and to NFPA 10.

Large chemical quantities delivered by hose line discharge can be supplied by stationary or mobile extinguishers having capacities of 500 pounds (225 kg), 1000 pounds (455 kg), and 2000 pounds (909 kg). Several hose line stations may be equipped with these extinguishers to protect a given area.

Extinguishers designed for stationary use may also be mounted on vehicles for the protection of larger subdivided areas. Manufacturers have made available specially-designed fire trucks with dry chemical capacities to 3000 pounds (1365 kg). The chemical may be discharged through hose lines, or through high-capacity turret nozzles that have a protection range of approximately 100 feet (30 meters). These fire trucks

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may also be equipped with supplementary water extinguishing equipment. However, when using large wheeled units, fire visibility may be lost because of the dust cloud.

4.5 COMBINED AGENTS

Since aqueous film-forming foam (AFFF) can be combined with dry chemicals, systems have been developed for simultaneous or alternate application of these foams and dry chemicals. Refer to NFPA 11 for system design criteria.

4.6 HALON EXTINGUISHING AGENTS

Halon extinguishing agents, such as 1211, 1301, and 2402, are no longer recommended for use in new installations in accordance with the Montreal Protocol, due to concerns regarding effects on the earth's ozone layer. Acceptable Halon replacement agents are currently available; *see NFPA 2001, Clean Agent Fire Extinguishing Systems. For maintenance of existing Halon systems, refer to NFPA 12A for Halon 1301; and NFPA 12B for Halon 1211 systems.*

4.7 CARBON DIOXIDE

An inert gas such as carbon dioxide ($C0_2$), discharged into a closed room or into enclosed spaces, can be effective in extinguishing fires in petroleum pump rooms, electrical installations, and for some special machinery or laboratory apparatus. Nevertheless, the asphyxiation hazard to personnel must be recognized and addressed in the use of an inert gas system. Due to static electricity hazard, carbon dioxide systems should not be used to inert a flammable atmosphere to prevent ignition. See NFPA 12 for additional information.

4.8 STEAM SMOTHERING

The general use of steam as an extinguishing agent can be ineffective. A substantial delay will occur before sufficient air is displaced or diluted to render the atmosphere incapable of supporting combustion. Steam is effective in special situations:

a. Smothering steam in furnace fire boxes and header boxes.
b. Steam rings on equipment flanges.
c. Steam rings on hot-tap equipment.

Steam should not be injected into large vapor spaces such as cone-roof tanks containing flammable mixtures; static electricity generation from such application is believed to have been the source of ignition for fires in the past.

5 Operating Practices

5.1 GENERAL

Standard Operations Procedures (SOPs) and Emergency Operating Procedures (EOPs) should be developed appropriate to facility needs and consistent with API RP 750 and regulatory requirements such as those in 29 *CFR* 1910.110, 1910.119 and 1910.156. Job and task procedures also should be developed. Supervisory and processing unit personnel should review SOPs and EOPs to assure they are both accurate and up-to-date and that procedures are being adhered to as written and intended. Supervision should inform employees of their roles and responsibilities for safe unit operation and the potential process consequences or hazards associated with deviation from SOPs and EOPs.

All personnel should understand their responsibility and accountability for safe unit operation and how unsafe or abnormal conditions or circumstances should be handled through appropriate levels of management. Appropriate action should be taken for abnormal circumstances, such as:

a. Leaks, odors, or unusual sound levels.
b. Accumulations of flammable liquids or gases.
c. Defective or damaged equipment.
d. Excessively high or low temperatures or pressures.
e. Failure to implement Management of Change procedures when changes to the process or equipment are made without appropriate management review and approval.
f. Infractions of safety, health and/or environmental regulations or procedures.
g. Unauthorized hot work.
h. Unauthorized vehicles or personnel in an area.
i. Malfunction of instrumentation or control systems that may affect safe unit operations.
j. Evidence of any structural weaknesses caused by deterioration.

A clear set of expectations should be developed so that personnel have clear knowledge of their role in all phases of the operation.

5.2 NORMAL OPERATIONS

The fire/loss prevention measures described in this section shall be considered for normal unit operation.

5.2.1 Operating procedures for furnaces, heaters, and other fired equipment should address purging, initial light off, re-ignition, draft controls, burner management systems, safety interlock systems, and fuel shut-off systems.

5.2.2 Locomotives, other than the fireless steam type, should not be permitted to operate near low flash point oil loading racks while cars are being loaded or while the area is contaminated with flammable vapors.

5.2.3 Operating procedures for loading tank trucks, tank cars, tankers, and barges should be prepared with review and consideration of the appropriate recommended precautions from API RP 2003.

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5.2.4 Grounding or electrical insulation of electrical equipment, lines, cargo hoses, and equipment shall follow API guidelines (see API RP 2003) for loading installations at docks, wharves, piers, loading terminals, and railway loading installations for dissipation of static charges.

5.2.5 Grounding or electrical isolation of hydrocarbon processing equipment should meet API RP 500. An assurance program to verify electrical grounding or isolation should be performed routinely.

5.2.6 Housekeeping procedures should be followed in all areas to prevent accumulations of oil, grease, or Class A materials (rags, wood, cardboard).

5.2.7 Fire and safety systems should be easily identified (color coded and identified), inspected routinely for operability, and maintained and calibrated in accordance with manufacturers' recommendations to assure they will function as designed and intended. Personnel expected to operate this equipment as part of their job responsibility should be trained on the equipment, its capabilities, and limitations.

5.2.8 Storage of flammable or combustible liquids should be done in accordance with NFPA 30.

5.2.9 SOPs for material transfers should be incorporated into daily operations. Examples include the following:

a. Tank gauging.
b. Tank sampling.
c. Tank transfer line up.
d. Tank filling and product movement.
e. Water draws.
f. Tank dike drain valve position.

5.2.10 A clear understanding of utility system capabilities, capacities, and limitations is needed by operating personnel to ensure safe operation.

5.2.11 Important lines and equipment should be identified.

5.2.12 Safe work permits are typically needed for the following type jobs:

a. Hot work.
b. Cold work.
c. Confined space entry.
d. Vehicle access.
e. Equipment isolation.

5.2.13 Control and maintenance of purge and pressurization systems for equipment that does not meet the electrical classification for the area is essential for control and suppression of ignition sources. See NFPA 496, *Purged and Pressurized Enclosures for Electrical Equipment in Hazardous (Classified) Locations.*

5.2.14 Precautions should be taken to exclude air and water from processes for which they were neither designed nor intended.

5.2.15 The addition or deletion of process chemicals such as demulsifiers and corrosion inhibitors should be done with the thorough understanding of metallurgical implications. Likewise, changes to operating parameters (e.g., transition temperatures), feed-stocks (particularly those containing sulfur), and equipment repairs involving hot work (normalizing, annealment, heat treatment, and weld repair) typically involve detailed job procedures that address metallurgical concerns.

5.2.16 Electrical junction boxes, receptacles, or other service devices should not be used in a classified area unless all bolts, covers, screws, and seals are maintained according to the manufacturers' recommendations and NFPA 70.

5.2.17 Temporary repairs for the control of hydrocarbon leaks such as epoxy injections, containment boxes (clamps), etc., should be engineered and have proper management review and approval. All such repairs should be considered temporary until such time as they are safely and permanently repaired (see API RP 570).

5.3 EMERGENCY OPERATIONS

The potential for fires, explosions, or gas releases increase during times of abnormal or emergency operation. Properly executed pre-incident plans responding to unusual circumstances are needed to address these conditions.

5.3.1 SOPs and EOPs should address job responsibilities and duties for unit personnel who will participate in startup and planned or emergency shutdowns of the process. These responsibilities should address the requirements described in API RP 750 and 29 *CFR* 1910.119.

5.3.2 Operating personnel should be informed of relief system and flare system capabilities and limitations to support safe operation under abnormal or adverse operations.

5.3.3 Operating personnel should be trained to develop awareness of the consequences of potential abnormal operations.

5.3.4 Management systems should be instituted to quickly notify potentially affected parties (nearby process units, adjacent facilities, and community neighbors). See Section 7.4.

5.3.5 EOPs should address abnormal events such as floods, hurricanes, tornadoes, earthquakes, high winds, lightning, heavy snows and ice storms, freezing temperatures, civil unrest and terrorism, where applicable. See Section 7 on "Fire Protection Organization."

5.3.6 Unit personnel must be informed of their emergency response (E/R) roles in the facility emergency management program. Personnel responses shall be consistent with OSHA

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29 *CFR* 1910.120 and 1910.156. Personnel assigned E/R roles should have appropriate levels of training and emergency equipment to safely carry out those roles and responsibilities. E/R drills should be a routine part of unit SOPs and EOPs. See Section 8.2.3, on "Simulated Fire Drills."

5.4 LEAKS

Since containment of hydrocarbons is the primary and most effective fire prevention principle, leaks are of significant concern. Considerations for the control of a leak should include the following:

a. Personnel exposure.
b. Utilization of emergency response personnel and resources.
c. Isolation of the fuel leak at the upstream source.
d. Isolation of transfer medium.
e. Isolation of ignition sources.
f. Containment of product.
g. Downwind and off-site impact.
h. Displacement of liquids still at risk.
i. Reduction of hazard zone via application of fire-fighting foam for vapor suppression.
j. Development of mitigation clean-up strategies.

5.4.1 Liquid Leaks

If a break or serious leak occurs in a line, the pumps should be shut down and all appropriate block valves closed. If possible, suction can be applied to the affected portion of the line unless the resultant entrance of air creates a hazard. In that case, water or inert gas displacement may be desirable. If the leak involves a tank or any large vessel, available lines should be used to pump out the liquid. Trenches, dikes, or diversion walls should be used either to confine the spill or to divert it to sewers or separators. If the leak is otherwise unstoppable, water may be used to displace the liquid.

If a large spill occurs, portable pumps may be required to supplement permanent equipment in recovering the hydrocarbon. If flammable vapors are present, caution must be used in positioning the pumps because the pump drivers may be sources of ignition and personnel exposure may be a concern. Traffic should be controlled and vehicles should be excluded from the affected area. Proper bonding and grounding should be used. Vent hoses from vacuum trucks must be placed with caution, to prevent ignition of the vapors (see API Publ 2219, *Safe Operating Guidelines for Vacuum Trucks in Petroleum Service*).

Water spray or steam applied at the emission point of a small leak may aid in dispersing vapors and preventing ignition. Foam may be applied to cover hydrocarbon spills, in order to reduce vaporization.

5.4.2 Gas Leaks

In the event of a break or failure of a hydrocarbon vapor or liquefied petroleum gas line or vessel, all nearby or downwind sources of ignition should be immediately eliminated. The vapors from leaks may roll along the ground and accumulate in low-lying areas before diluting and dispersing. Large leaks have the potential to travel great distances and still be within the flammable limits for hydrocarbon. Some gas leaks may be toxic to humans, as well as being flammable; emergency responders should use appropriate personal protective equipment and people downwind may need to shelter-in-place. It may be possible to disperse vapor leaks by utilizing air, steam, water or other agents to mitigate the potential flammable or toxic hazards. If ignition occurs, a pressure fire should never be extinguished until the source of the leak can be identified and isolated.

6 Maintenance Procedures

6.1 GENERAL

Fire prevention during maintenance activities depends on careful planning and preparation with isolation or the removal of flammable liquids, vapors, sludges or other materials before any work is begun. Related major activities such as demolition or the removal of process equipment poses unique hazards. Lines and equipment must be identified and marked. Hydrocarbon-freeing procedures must be followed, such as blinding, purging, gas freeing, steaming, water washing, liquid evacuation, or inerting.

In normal maintenance and during demolition it is sometimes impossible to verify the complete removal of flammable material or elimination of oxidizers. Therefore, precautions must be taken to prevent ignition sources from contacting flammable mixtures. Examples include the following:

a. Isolation of drainage in the work area.
b. Inerting containers.
c. Enriching containers.
d. Using spark containment enclosures around the work area.
e. Use of alternatives to hot work (cold cutting, hydraulic scissors cuts, using pneumatic tools).

Good housekeeping is an important prerequisite in preventing fires associated with maintenance procedures. Refuse that is combustible should be stored in appropriate containers. Leaks and spills should be addressed promptly.

6.2 HOT WORK

Alternatives to the use of hot work in process areas should be considered when work must be conducted away from designated shop areas. However, hot work activities such as electric or gas welding, cutting, brazing, or similar flame or spark-producing operations are necessary for certain equipment maintenance at refineries. Hot work maintenance activities normally require a hot work permit or authorization provided by experienced or trained individuals. These hot work activities may be subject to several regulations, for instance OSHA 29 *CFR* 1910.119, 1910.147 and 1910.252. Historically, U.S. Coast

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Guard permit requirements have been an additional consideration for hot work at docks associated with refineries. For more information see API Publication 2009.

Certain maintenance activities may require welding or hot tapping on equipment in service if it is infeasible or impractical to take the equipment out of service. Such activities require careful planning. For additional information see API Recommended Practice 2201.

6.3 PLANNED MAINTENANCE ACTIVITIES

The following elements should be considered for inclusion in planned maintenance activities:

a. Written maintenance procedures.
b. Up-to-date engineering, drawings, inspection procedures and reports, manufacturers information, and historical information on repairs.
c. Positive Material Identification (PMI) programs.
d. Written safe work authorization procedures.
e. Established rigging and lifting procedures.
f. Use of tools and equipment (see API 2214).
g. Adherence to grounding and bonding procedures, with static discharge awareness (see API 2003).
h. Control of hazardous energy and material sources (lockout/tagout, see OSHA 29 *CFR* 1910.147).
i. Procedures for bypassing critical shutdown and safety equipment and their subsequent return to service.
j. Maintenance inspection of pressure vessels (see API 510 and API RP 572).
k. Piping inspection and repair (see API 570).
l. Tank inspection and repair (see API Publ 653, RP 575, and Std 2610).
m. Management of Change Procedures (see API 750 and OSHA 29 *CFR* 1910.119).

6.4 WINTERIZING

Prevention programs in areas with freeze potential should verify that out-of-service piping is freeze-proof; and provide for winterizing to reduce potential for ice formation and fracture of aboveground fire water equipment, safety showers, and dead-leg pockets in which water can accumulate (such as bottom gauge glasses, low spots in pipe runs, draw connections, bleeders, and valve manifolds).

7 The Fire Protection Organization

7.1 GENERAL

This section reviews the basic principles of fire-protection organization with the recognition that many different types of organizations function effectively.

7.2 INCIDENT COMMAND SYSTEM

The Incident Command System (ICS) is an integrated management system for emergencies such as fires, hazardous material spills, multicasualty incidents, earthquakes, floods, etc. ICS provides a management structure with defined modular roles for coordination of facility personnel and operations, local fire departments, mutual aid organizations, and equipment responding to an emergency. For ICS to function effectively, training and education of all personnel (including management) who will assume ICS roles is important. Use of ICS is mandated by OSHA in *Hazardous Waste Operations and Emergency Response* 1910.120(q). Additional information on ICS is available in publications from the Federal Emergency Management Administration (FEMA)[13], the National Fire Academy,[14] Oklahoma State University's[15] Fire Protection Publications and in NFPA 1561 which is primarily directed toward public fire departments.

Many refineries designate facilities and staff to function as an Emergency Operations Center (EOC). The EOC addresses incident related management needs not specific to the incident mitigation strategy or tactics. These include internal interface with noninvolved operational units and external relations with the community, regulatory agencies and the media. Incident Command may operate from the same location as the EOC but frequently is situated in close proximity to the incident site while maintaining close communication with the EOC. This provides functional advantages for communication with emergency response and process operations personnel as well as useful isolation from distraction.

Firefighting is only one aspect of handling fire incidents. An ICS incorporating an EOC can be used to manage a wide range of related emergency activities which must be coordinated with firefighting, which can include:

a. Implementing emergency action and evacuation plans.
b. Accounting for personnel from the area affected by the incident.
c. Providing rescue and first aid for the injured.
d. Shutting down equipment and rerouting fuel from the fire area.
e. Performing special emergency maintenance work.
f. Controlling utilities.
g. Providing auxiliary traffic control and security.
h. Shutting off nonessential water usage from the fire main when a fire alarm is activated.
i. Transporting and staging reserve personnel and fire-fighting equipment.
j. Maintaining a system to account for personnel working in the "hot zone."
k. Providing rehabilitation areas for response personnel.
l. Ensuring liaison among all the emergency activities.

[13]Federal Emergency Management Agency, 500 C Street, S.W., Washington, D.C., 20472. www.fema.gov
[14]National Fire Academy, 16825 South Seton Avenue, Emmittsburg, Maryland 21727-8998. www.usfa.fema.gov
[15]Oklahoma State University, Stillwater, Oklahoma 74078. www.osu.okstate.edu

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m. Providing for media communications and good public relations.

n. Providing for backup operating personnel.

o. Making mandatory timely notifications of federal, state, or local agencies.

Normally the person responsible for the unit where the fire has occurred should take the lead role for emergency shutdown operations. An alternate should be assigned in case the responsible person is not available. Supervision of the actual firefighting is the responsibility of the Incident Commander or a designated alternate. Coordination between the persons in these two roles is a key to safe and effective fire control.

7.3 DUTIES OF FIRE PROTECTION STAFF

Because of variables among refineries (such as size, management structure, available personnel, throughput, nature of operations, philosophy of fire control, and facilities used), each refinery may choose a unique type of fire-protection organization to meet its particular needs. In many refineries the fire protection organization consists of a staff who may supervise fire-fighting activities, conduct fire training, inspect emergency response equipment, participate in rescue and emergency medical activities, maintain and issue fire-protection equipment, and investigate and report fires. This staff may also counsel operating and engineering groups on fire protection for new or rebuilt facilities, attend planning sessions for major shutdowns of operating equipment, work with fire insurance representatives, and confer with regulatory agencies on codes and ordinances. In some refineries, this staff is integrated with the safety or accident-prevention group. In many refineries a volunteer fire organization operates effectively to fulfill a portion of the above duties.

Periodic review of a refinery's fire protection program is recommended so that the fire-fighting organization will function effectively. Written policy, procedures, and training requirements are specified in OSHA *CFR* 29 1910.156 and 1910.120(q).

7.4 NOTIFICATION PROCEDURES

A callout notification system should be developed to contact off-site personnel with emergency response duties, including those with Incident Command or Emergency Operations Center roles.

When a fire is reported, the procedure should include prompt notification of the facility fire fighting and emergency team personnel, outside community or regulatory authorities requiring notification, and any outside mutual aid groups who may be called upon to provide equipment, supplies, or personnel. Procedures should be in place to establish sequence and priorities for notification depending upon incident needs.

In some refineries, arrangements are made for the fire or emergency calls to be received at constantly-attended locations, such as a laboratory, powerhouse, or main gate. Special telephones are used only for incoming emergency calls. A dedicated phone number is selected, and decals showing this number are attached to all in-plant refinery phones.

The alarm procedure is initiated by the attendant receiving the emergency call. The attendant should be trained for the duty and be supplied in advance with the following:

a. An emergency call-out list for key personnel, local public fire departments, ambulance services, and doctors. In some instances, an independent agency is used for handling this emergency call-out.

b. A set of written notification scripts specific to incident type and personnel or agency being notified.

c. An emergency communication system between the main office, the main gate, and other key locations. The location and nature of the emergency will be announced over this system.

d. A method for recording all calls (e.g., log book, tape recorder, etc.) and time of notification.

e. A checklist to ensure that all notifications appropriate to the type of incident have been completed.

Issuance of company identification to key people who may have to respond from offsite (including members of the fire-fighting squads, Incident Command and EOC personnel) is desirable in order to assist passage through road blocks when proceeding to the refinery during an emergency. Firefighters should wear personal and organization identification on their clothing or headgear.

7.5 FIRE-FIGHTER SELECTION AND TRAINING

In developing a fire-fighting organization, the number of firefighters should be appropriate to the anticipated need. Specific types of personnel should be chosen for the firefighting organization. Selection should be based on the number of workers available, their job assignments, diversity of skills, their freedom to be relieved of duty to respond to emergencies, and the type of fire protection equipment available. Care should be taken to verify that fire-fighting personnel are physically capable of performing the duties that are assigned.

A core group of firefighters should be selected from regular shift employees who can be relieved from regular duties for firefighting; they, in turn, should be supplemented by day staff or off-duty employees. Some locations select shift personnel for fire-fighting duties on the basis of job assignment or classification. Other facilities use a voluntary basis, rather than assigning fire-fighting duties to specific individuals within the plant. Many plants use a combination of assigned and volunteer fire crews. Consideration should be given to shift changes, days off-duty, and vacations to provide adequate coverage at all times. Where possible, emergency plans

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should use supplemental assistance from outside fire departments or mutual-aid groups for major incidents.

When working with volunteer or other supplemental fire-fighting groups, it is important to determine which personnel will be in charge of each aspect of the fire-fighting activity. Use of the ICS provides an excellent functional tool for this coordination and satisfies regulatory requirements.

Most fires have the potential for the emergency response action to include shutting down equipment, as well as fighting the fire. The division of responsibilities among emergency activities should be made through a pre-fire training program. Emergency simulations and unit specific drills help prepare for fire incidents. The training program in Section 8 outlines one approach. It should be tailored to apply to a particular refinery or to the needs of specific types of potential incidents. Training should be structured to develop an effective emergency response capability as well as provide regulatory compliance.

7.6 THE INCIDENT COMMANDER

Operating under the ICS one individual should be designated as the Incident Commander (IC). This individual should have received appropriate ICS training. The IC role is modular and can be passed to another individual as personnel availability changes. The Incident Command System structure provides an expandable framework to address both small and large incidents Many locations use ICS for all incidents to provide a consistent approach and maintain familiarity with the process and terminology.

The Incident Commander should have multiple reliable means of communication so that, during an emergency, he or she can promptly contact those involved in fire control operations. Portable radios, cellular phones, and plant phones are some options. Communication with potential mutual aid responders should be reviewed in advance both for incident notification and for subsequent on-site incident management. A person who is fully conversant with radio-telephone equipment can be of great value to coordinate the fire emergency communications with the regular plant communications system. The type of communication used will be determined by the refinery's communications resources, as well as by the type and size of the fire.

7.7 FIRE-FIGHTER PERSONAL PROTECTIVE CLOTHING AND EQUIPMENT

7.7.1 Protective Clothing

Proper protective clothing is required for fire brigade members and emergency responders based on their anticipated exposure and duties. Federal regulatory requirements are outlined in OSHA 29 *CFR* 1910.156 and 1910.120(q). NFPA 600 *Industrial Fire Brigades* states that clothing shall be available in sufficient quantity and sizes to fit each brigade member expected to fight advanced exterior and/or interior structure fires. NFPA 600 states that all firefighters' protective clothing (frequently called bunker gear or turnouts) must meet the requirements of the appropriate NFPA standard, (see NPFA 1971, 1972, 1973, and 1974). Non-fire brigade personnel working in a fire-exposed area should be similarly protected.

7.7.2 Respiratory Protection And Other Equipment

Positive pressure self contained breathing apparatus (SCBA) shall be used by potentially exposed fire brigade personnel when, in the judgment of the personnel in charge (i.e., Incident Commander, Safety Officer, Fire Chief, etc.) the atmosphere is hazardous, suspected of being hazardous, or could rapidly become hazardous. Other equipment such as Personal Alert Safety Systems (PASS) devices should be used when appropriate, such as when fighting interior structural fires.

8 Training for Firefighting

8.1 GENERAL

The primary goal for emergency response training is to develop and maintain an effective organization which functions safely during emergency situations without unduly putting individuals at risk. Written training policies and procedures for firefighting should be developed. All personnel should receive training commensurate with level of responsibility and which prepares them for the duties they will be assigned. This section discusses some of the various types of fire training currently in use. Refer to OSHA 29 *CFR* 1910, Subpart L and 1910.120(q) for minimum regulatory training requirements, and NFPA 600 for guidance. Instructors should be familiar with the characteristics and limitations of the equipment used. Training should be conducted both in the classroom and on the drill grounds, using various types and sizes of live fires and simulations. A system for tracking individual training should be used; credit for training received through other venues (such as volunteer fire departments or individual study course work) can be considered for maintaining current status.

8.2 DRILL GROUND TRAINING

When "hands-on" training is conducted on-site, drill grounds may be built to provide adequate training for the facility involved. Training should not be limited solely to pit, tank, and other two-dimensional fires. Realistic props can be constructed from scrap equipment and can be used to simulate jet, spray, and other three-dimensional pressure fires potentially encountered on the job. Where possible, more than one type of fuel should be used for suppression training to build experience and skills which reflect the range of materials likely to be encountered in fires at the location. Training

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plans should give consideration to providing experience with hydrocarbon fires for personnel from public departments who may respond as mutual aid. Onsite training can be supplemented using offsite training from fire training schools available throughout the country.

8.2.1 Instructors

Instructors and those expected to lead firefighters must be provided with training that is more comprehensive than that provided to the general membership of the fire crew. Qualified drill ground instructors may come from the plant fire brigade, plant staff, or outside personnel; they may also be plant personnel, trained by skilled instructors, who in turn train the staff working with them. The qualifications of all instructors should be documented.

8.2.2 Types of Training

Personnel expected to respond to fires should be provided with training consistent with those expectations. This should include primary education or training for whatever equipment they are expected to use.

For all responders, including those with incipient firefighting roles, this typically will include identifying the need for, and use of, personal protective equipment and facilities designed for use by one person (such as hand extinguishers, small water and foam hose lines) as well as activation of fixed water-spray systems, fixed monitors, and deluge sets.

Secondary training for brigade members may include handling of equipment such as large hose lines, fire mains and foam equipment, fire apparatus, self-contained breathing apparatus (SCBA), and related equipment. Training activities should strive to develop teamwork among the various groups assigned to respond to major fires.

Training of outside emergency responders is also important. If public or other outside fire departments respond to plant fire alarms, it may be advantageous for their personnel (especially officers) to participate in some of the training used for the refinery emergency responders.

8.2.3 Simulated Fire Drills

Simulated fire drills should be used in a facility's firefighting training program. In large facilities these may be related to unit-specific pre-fire incident plans (see Section 9.2). On-the-job simulated emergency (hypothetical) fire drills relate the drill-ground training to potential emergency problems in operating areas. This can be achieved if operating personnel and emergency response personnel jointly plan a simulated emergency. The drill should activate those portions of the ICS which would be incorporated in a "real" response.

For example, a stubborn fire from a blown packing gland may be simulated. If the unit is operating smoothly, appropriate personnel should be advised of the drill, notifications made and the alarm should be sounded. The firefighters should then respond with the appropriate fire-fighting equipment and Incident Command should be established. Fire suppression setup should proceed while the operating personnel simulate emergency shutdown operations. These shutdown operations may be accomplished by placing tags or small fiber gaskets, marked "open" or "closed," on valves, pumps, and the like. This approach is sometimes called a "red tag" drill.

For observation purposes, one or more persons may be assigned to keep a record of the time required to activate fire-extinguishing equipment and to report on the overall efficiency of the fire drill. The lessons learned from the drill should then be discussed with the operators, and, if necessary, the drill may be repeated to improve performance. At the conclusion of the drill, a critique should be conducted to evaluate the effectiveness of the response and how it might be improved.

Strategies should be developed to direct the control and extinguishment of various types of refinery fires. Strategies may include evaluating the means to provide appropriate resources to resolve an overall problem, identifying specific needs such as lifesaving and rescue techniques, and covering exposures. Drills using these strategies should be conducted periodically in various sections of the refinery.

If feasible, drills should be expanded to include the activation of spray systems and fixed monitors and the use of large hose lays. The appropriate fire crews should be employed in these drills.

8.3 CLASSROOM INSTRUCTION

Basic classroom instruction can include discussion of firefighter safety, table-top demonstrations, fire tetrahedron theory including the burning characteristics of various fuels, and education regarding the function of the various types of fire-fighting agents used at the facility. This classroom instruction should include a review of:

a. Safety procedures for emergency responders.
b. Personal protective equipment for firefighters.
c. Practice in the use of self-contained breathing apparatuses.
d. ICS operation and roles.
e. Fire-related special hazards in the workplace.
f. Review of any special hazardous materials showing quantities and location.
g. Emergency action procedures.
h. Pre-fire plan.
i. Fire-fighting tactics.
j. Fire water and foam systems.

Study of past fires is an important aspect of the fire training program and should be part of the committee agenda. When possible, reports of fires in other refineries should be studied to increase knowledge of fire-fighting problems specific to refineries.

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8.4 OVERCOMING PERSONAL CONCERNS

Many of the psychological reactions and concerns experienced by individuals when they face a large fire may be overcome with proper drill ground training. Persons trained on large fires recognize the magnitude of their task and usually do not become overly confident. They also learn the limitations of their equipment.

8.5 DOCUMENTATION

Training records should be maintained to ensure readiness of the emergency response personnel and document regulatory compliance. Written training requirements based on emergency responder duties and roles are specified in the Occupational Safety and Health Act, Subpart L Fire Brigades (1910.156), and under Hazardous Waste Operations and Emergency Response 1910.120(q).

9 Pre-Fire Plan

9.1 GENERAL

To effectively coordinate the activities essential to pre-fire planning, some facilities have developed a written Pre-Fire Plan which addresses these issues. This plan is designed for a major facility, but the basic principles apply to all facilities. The plan may need to be tailored to the size and complexity of the specific organization. The plan should follow the existing departmental organization, to avoid disruption of normal operations.

9.2 PRE-FIRE INCIDENT PLANNING

Pre-fire incident plans provide effective tools for reviewing response capability and for structuring training. They also assist in regulatory compliance (as for 29 *CFR* 1910.38). Pre-fire plans should provide guidance for addressing special concerns such as BLEVE, boilover, or water reactive chemicals.

Candidate subject areas for inclusion in a Pre-Fire Plan might include:

a. Possible fire hazards including review of hazardous material inventory.

b. Water supply availability.

c. Water requirements (extinguishing, cooling, etc.).

d. Foam delivery requirements and capability.

e. Typical weather concerns.

f. Response requirements (e.g., personnel requirements, delivery equipment, and consumables such as foam, dry chemical, carbon dioxide, etc.)

g. Mutual aid organization capabilities, resources and response time.

h. Evacuation requirements.

i. Communication needs.

j. Scaled plot plan of the hazard and/or areas potentially involved.

k. Scene security.

l. Accessibility to scene (e.g., having an alternate route in case access to the scene is blocked by rail cars or other equipment).

m. Decontamination facilities and procedures.

n. Designation of staging areas.

o. Major medical response.

p. Industrial rescue capability.

q. Areas with asbestos.

r. Location of radioactive instrumentation elements.

s. Potential for polychlorinated biphenols (PCBs).

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